

SIMULATION IDEAS

The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right).$$

Case 1: 2D Tensor Covariate

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p'_1 = p'_2 = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathbf{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\begin{aligned} \alpha_{1rs}(t) &= 5\sqrt{\frac{rs}{RS}}(t - 0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} - 0.75 \right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)), \\ \alpha_{4rs}(t) &= \sqrt{\frac{rs}{RS}} \left[0.45 \cdot 5(t - 0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20 \sin(2\pi(t - 0.5)) \right], \\ \alpha_{5rs}(t) &= \sqrt{\frac{rs}{RS}} \left[1.10 \exp \left\{ -\frac{(t - 0.30)^2}{2(0.08)^2} \right\} - 0.95 \exp \left\{ -\frac{(t - 0.72)^2}{2(0.11)^2} \right\} \right], \\ \alpha_{6rs}(t) &= \sqrt{\frac{rs}{RS}} \left[18(t - 0.2)(t - 0.55)(t - 0.85) - 0.05 \right] \end{aligned}$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. We will vary $\sigma^2(t)$ to 4 subcases as Case 1(a)-(d).

$$\sigma_0^2(t) \equiv 1, \quad \sigma_1^2(t) = 1 + 0.3 \sin(2\pi t), \quad \sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t), \quad \text{and} \quad \sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t). \quad (2)$$

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) in AOAS draft.

Simulation Results. We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Table 1: MAE errors ($\times 10^{-2}$) for Case 1(a) - $\sigma_0(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.06 ($\pm$ 0.17)	5.08 ($\pm$ 0.18)	2.09 ($\pm$ 0.78)	2.12 ($\pm$ 0.77)	30.32 ($\pm$ 2.66)	2.81 ($\pm$ 1.91)
		0.3	5.05 ($\pm$ 0.20)	4.94 ($\pm$ 0.18)	2.44 ($\pm$ 1.01)	2.49 ($\pm$ 1.07)	30.23 ($\pm$ 3.48)	7.48 ($\pm$ 3.06)
		0.6	5.06 ($\pm$ 0.24)	4.53 ($\pm$ 0.17)	2.66 ($\pm$ 1.10)	2.69 ($\pm$ 1.13)	30.19 ($\pm$ 3.71)	15.09 ($\pm$ 2.55)
		0.9	5.06 ($\pm$ 0.27)	3.57 ($\pm$ 0.18)	2.88 ($\pm$ 1.13)	2.84 ($\pm$ 1.15)	30.14 ($\pm$ 3.99)	22.70 ($\pm$ 1.92)
	2000	0.0	3.29 ($\pm$ 0.10)	3.30 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.24 ($\pm$ 0.44)	13.31 ($\pm$ 1.88)	1.37 ($\pm$ 0.80)
		0.3	3.32 ($\pm$ 0.11)	3.22 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.36 ($\pm$ 0.61)	13.24 ($\pm$ 1.82)	4.12 ($\pm$ 1.64)
		0.6	3.33 ($\pm$ 0.11)	2.89 ($\pm$ 0.10)	1.53 ($\pm$ 0.65)	1.47 ($\pm$ 0.67)	13.15 ($\pm$ 2.04)	8.48 ($\pm$ 1.41)
		0.9	3.33 ($\pm$ 0.11)	2.16 ($\pm$ 0.07)	1.68 ($\pm$ 0.70)	1.56 ($\pm$ 0.72)	13.07 ($\pm$ 2.32)	12.92 ($\pm$ 1.03)
$\mathbf{A}_2(t)$	1000	0.0	5.08 ($\pm$ 0.21)	5.10 ($\pm$ 0.21)	2.09 ($\pm$ 0.79)	2.12 ($\pm$ 0.78)	30.33 ($\pm$ 2.72)	2.94 ($\pm$ 2.11)
		0.3	5.03 ($\pm$ 0.18)	4.92 ($\pm$ 0.16)	2.44 ($\pm$ 0.98)	2.51 ($\pm$ 1.03)	30.28 ($\pm$ 3.56)	6.87 ($\pm$ 2.92)
		0.6	5.03 ($\pm$ 0.20)	4.50 ($\pm$ 0.16)	2.66 ($\pm$ 1.08)	2.71 ($\pm$ 1.11)	30.25 ($\pm$ 3.78)	14.46 ($\pm$ 2.31)
		0.9	5.02 ($\pm$ 0.22)	3.52 ($\pm$ 0.15)	2.89 ($\pm$ 1.12)	2.84 ($\pm$ 1.14)	30.20 ($\pm$ 4.05)	22.09 ($\pm$ 1.48)
	2000	0.0	3.28 ($\pm$ 0.08)	3.29 ($\pm$ 0.08)	1.21 ($\pm$ 0.46)	1.21 ($\pm$ 0.46)	13.43 ($\pm$ 1.89)	1.56 ($\pm$ 0.96)
		0.3	3.29 ($\pm$ 0.12)	3.19 ($\pm$ 0.11)	1.39 ($\pm$ 0.59)	1.37 ($\pm$ 0.63)	13.34 ($\pm$ 1.78)	3.87 ($\pm$ 1.59)
		0.6	3.29 ($\pm$ 0.12)	2.86 ($\pm$ 0.10)	1.54 ($\pm$ 0.64)	1.48 ($\pm$ 0.70)	13.24 ($\pm$ 2.03)	8.22 ($\pm$ 1.31)
		0.9	3.29 ($\pm$ 0.11)	2.09 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.60 ($\pm$ 0.74)	13.12 ($\pm$ 2.34)	12.64 ($\pm$ 0.98)
$\mathbf{A}_3(t)$	1000	0.0	6.49 ($\pm$ 0.22)	6.60 ($\pm$ 0.23)	2.21 ($\pm$ 0.76)	2.27 ($\pm$ 0.76)	17.16 ($\pm$ 2.64)	2.85 ($\pm$ 2.21)
		0.3	6.52 ($\pm$ 0.18)	6.56 ($\pm$ 0.17)	2.46 ($\pm$ 1.11)	2.53 ($\pm$ 1.09)	17.08 ($\pm$ 3.21)	11.00 ($\pm$ 3.08)
		0.6	6.53 ($\pm$ 0.19)	6.29 ($\pm$ 0.16)	2.70 ($\pm$ 1.18)	2.73 ($\pm$ 1.16)	17.15 ($\pm$ 3.26)	22.32 ($\pm$ 2.77)
		0.9	6.54 ($\pm$ 0.22)	5.75 ($\pm$ 0.15)	2.95 ($\pm$ 1.21)	2.89 ($\pm$ 1.22)	17.24 ($\pm$ 3.35)	33.66 ($\pm$ 2.46)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.49 ($\pm$ 0.56)	1.48 ($\pm$ 0.58)	9.82 ($\pm$ 1.17)	1.33 ($\pm$ 0.98)
		0.3	5.05 ($\pm$ 0.12)	5.09 ($\pm$ 0.12)	1.41 ($\pm$ 0.59)	1.38 ($\pm$ 0.61)	10.29 ($\pm$ 1.30)	8.32 ($\pm$ 1.90)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.55 ($\pm$ 0.65)	1.49 ($\pm$ 0.66)	10.41 ($\pm$ 1.36)	16.81 ($\pm$ 1.83)
		0.9	5.06 ($\pm$ 0.13)	4.50 ($\pm$ 0.09)	1.67 ($\pm$ 0.70)	1.58 ($\pm$ 0.70)	10.57 ($\pm$ 1.48)	25.32 ($\pm$ 1.67)
$\mathbf{A}_4(t)$	1000	0.0	4.87 ($\pm$ 0.19)	4.88 ($\pm$ 0.20)	2.09 ($\pm$ 0.76)	2.13 ($\pm$ 0.75)	31.77 ($\pm$ 2.69)	2.85 ($\pm$ 1.97)
		0.3	4.85 ($\pm$ 0.18)	4.73 ($\pm$ 0.16)	2.43 ($\pm$ 1.01)	2.47 ($\pm$ 1.07)	31.68 ($\pm$ 3.47)	6.83 ($\pm$ 2.97)
		0.6	4.85 ($\pm$ 0.21)	4.28 ($\pm$ 0.15)	2.65 ($\pm$ 1.10)	2.67 ($\pm$ 1.14)	31.64 ($\pm$ 3.69)	13.98 ($\pm$ 2.37)
		0.9	4.86 ($\pm$ 0.24)	3.20 ($\pm$ 0.16)	2.88 ($\pm$ 1.13)	2.82 ($\pm$ 1.15)	31.59 ($\pm$ 3.97)	21.14 ($\pm$ 1.64)
	2000	0.0	3.02 ($\pm$ 0.09)	3.02 ($\pm$ 0.09)	1.26 ($\pm$ 0.45)	1.26 ($\pm$ 0.45)	15.12 ($\pm$ 1.87)	1.44 ($\pm$ 0.78)
		0.3	3.03 ($\pm$ 0.09)	2.91 ($\pm$ 0.09)	1.36 ($\pm$ 0.58)	1.33 ($\pm$ 0.60)	15.04 ($\pm$ 1.85)	3.57 ($\pm$ 1.45)
		0.6	3.04 ($\pm$ 0.09)	2.54 ($\pm$ 0.08)	1.51 ($\pm$ 0.64)	1.45 ($\pm$ 0.66)	14.95 ($\pm$ 2.06)	7.33 ($\pm$ 1.30)
		0.9	3.04 ($\pm$ 0.09)	1.63 ($\pm$ 0.06)	1.65 ($\pm$ 0.69)	1.54 ($\pm$ 0.71)	14.85 ($\pm$ 2.35)	11.21 ($\pm$ 0.88)
$\mathbf{A}_5(t)$	1000	0.0	7.88 ($\pm$ 0.22)	8.19 ($\pm$ 0.24)	2.56 ($\pm$ 1.05)	2.49 ($\pm$ 1.00)	28.32 ($\pm$ 2.37)	3.50 ($\pm$ 2.04)
		0.3	7.84 ($\pm$ 0.26)	8.09 ($\pm$ 0.26)	2.63 ($\pm$ 0.83)	2.71 ($\pm$ 0.96)	28.28 ($\pm$ 2.54)	14.55 ($\pm$ 3.56)
		0.6	7.85 ($\pm$ 0.27)	7.89 ($\pm$ 0.26)	2.85 ($\pm$ 0.88)	2.93 ($\pm$ 1.03)	28.21 ($\pm$ 2.58)	29.92 ($\pm$ 3.58)
		0.9	7.85 ($\pm$ 0.30)	7.48 ($\pm$ 0.28)	3.08 ($\pm$ 0.93)	3.13 ($\pm$ 1.11)	28.19 ($\pm$ 2.66)	45.30 ($\pm$ 3.57)
	2000	0.0	6.46 ($\pm$ 0.14)	6.77 ($\pm$ 0.14)	1.59 ($\pm$ 0.55)	1.53 ($\pm$ 0.50)	38.31 ($\pm$ 2.77)	2.00 ($\pm$ 1.43)
		0.3	6.48 ($\pm$ 0.13)	6.73 ($\pm$ 0.13)	1.66 ($\pm$ 0.63)	1.61 ($\pm$ 0.55)	37.98 ($\pm$ 2.08)	13.04 ($\pm$ 2.74)
		0.6	6.49 ($\pm$ 0.13)	6.57 ($\pm$ 0.13)	1.79 ($\pm$ 0.67)	1.74 ($\pm$ 0.59)	38.02 ($\pm$ 2.04)	26.38 ($\pm$ 2.59)
		0.9	6.48 ($\pm$ 0.13)	6.24 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.85 ($\pm$ 0.61)	38.11 ($\pm$ 2.09)	39.79 ($\pm$ 2.32)
$\mathbf{A}_6(t)$	1000	0.0	6.18 ($\pm$ 0.21)	6.26 ($\pm$ 0.21)	2.10 ($\pm$ 0.84)	2.15 ($\pm$ 0.81)	20.74 ($\pm$ 2.87)	2.91 ($\pm$ 2.08)
		0.3	6.13 ($\pm$ 0.21)	6.12 ($\pm$ 0.20)	2.62 ($\pm$ 1.01)	2.64 ($\pm$ 1.05)	20.63 ($\pm$ 3.74)	9.38 ($\pm$ 3.13)
		0.6	6.12 ($\pm$ 0.23)	5.81 ($\pm$ 0.20)	2.83 ($\pm$ 1.12)	2.82 ($\pm$ 1.13)	20.62 ($\pm$ 3.94)	19.66 ($\pm$ 2.69)
		0.9	6.11 ($\pm$ 0.25)	5.18 ($\pm$ 0.20)	3.04 ($\pm$ 1.19)	2.96 ($\pm$ 1.18)	20.62 ($\pm$ 4.18)	29.96 ($\pm$ 2.07)
	2000	0.0	4.59 ($\pm$ 0.10)	4.65 ($\pm$ 0.10)	1.17 ($\pm$ 0.45)	1.17 ($\pm$ 0.45)	6.90 ($\pm$ 1.09)	1.67 ($\pm$ 1.00)
		0.3	4.61 ($\pm$ 0.13)	4.61 ($\pm$ 0.12)	1.55 ($\pm$ 0.64)	1.54 ($\pm$ 0.67)	7.11 ($\pm$ 1.22)	6.86 ($\pm$ 1.85)
		0.6	4.61 ($\pm$ 0.13)	4.40 ($\pm$ 0.11)	1.71 ($\pm$ 0.69)	1.66 ($\pm$ 0.74)	7.16 ($\pm$ 1.31)	14.36 ($\pm$ 1.69)
		0.9	4.60 ($\pm$ 0.14)	3.95 ($\pm$ 0.08)	1.86 ($\pm$ 0.73)	1.79 ($\pm$ 0.78)	7.26 ($\pm$ 1.44)	21.94 ($\pm$ 1.52)

Table 2: MAE errors ($\times 10^{-2}$) for Case 1(b) - $\sigma_1(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.05 ($\pm$ 0.18)	5.05 ($\pm$ 0.18)	2.12 ($\pm$ 0.80)	2.05 ($\pm$ 0.75)	30.43 ($\pm$ 2.75)	2.93 ($\pm$ 1.94)
		0.3	5.03 ($\pm$ 0.20)	4.92 ($\pm$ 0.18)	2.41 ($\pm$ 1.03)	2.45 ($\pm$ 1.04)	30.27 ($\pm$ 3.47)	7.63 ($\pm$ 3.19)
		0.6	5.03 ($\pm$ 0.23)	4.51 ($\pm$ 0.17)	2.63 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	30.26 ($\pm$ 3.76)	15.23 ($\pm$ 2.67)
		0.9	5.04 ($\pm$ 0.27)	3.56 ($\pm$ 0.18)	2.86 ($\pm$ 1.14)	2.79 ($\pm$ 1.12)	30.23 ($\pm$ 4.12)	22.80 ($\pm$ 2.02)
	2000	0.0	3.29 ($\pm$ 0.10)	3.29 ($\pm$ 0.10)	1.24 ($\pm$ 0.46)	1.21 ($\pm$ 0.42)	13.47 ($\pm$ 1.92)	1.60 ($\pm$ 0.90)
		0.3	3.31 ($\pm$ 0.11)	3.20 ($\pm$ 0.11)	1.39 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	13.41 ($\pm$ 1.93)	4.31 ($\pm$ 1.55)
		0.6	3.32 ($\pm$ 0.11)	2.88 ($\pm$ 0.10)	1.53 ($\pm$ 0.67)	1.44 ($\pm$ 0.65)	13.46 ($\pm$ 2.10)	8.63 ($\pm$ 1.31)
		0.9	3.32 ($\pm$ 0.11)	2.15 ($\pm$ 0.07)	1.67 ($\pm$ 0.72)	1.53 ($\pm$ 0.69)	13.57 ($\pm$ 2.25)	12.97 ($\pm$ 0.98)
$\mathbf{A}_2(t)$	1000	0.0	5.05 ($\pm$ 0.21)	5.07 ($\pm$ 0.21)	2.11 ($\pm$ 0.82)	2.06 ($\pm$ 0.75)	30.43 ($\pm$ 2.82)	3.14 ($\pm$ 2.03)
		0.3	4.99 ($\pm$ 0.18)	4.89 ($\pm$ 0.17)	2.42 ($\pm$ 1.01)	2.44 ($\pm$ 1.02)	30.32 ($\pm$ 3.54)	6.97 ($\pm$ 3.07)
		0.6	4.99 ($\pm$ 0.20)	4.47 ($\pm$ 0.16)	2.66 ($\pm$ 1.09)	2.63 ($\pm$ 1.11)	30.32 ($\pm$ 3.83)	14.55 ($\pm$ 2.42)
		0.9	4.99 ($\pm$ 0.22)	3.49 ($\pm$ 0.15)	2.88 ($\pm$ 1.15)	2.78 ($\pm$ 1.12)	30.30 ($\pm$ 4.18)	22.12 ($\pm$ 1.51)
	2000	0.0	3.26 ($\pm$ 0.08)	3.27 ($\pm$ 0.08)	1.19 ($\pm$ 0.46)	1.18 ($\pm$ 0.44)	13.55 ($\pm$ 1.99)	1.77 ($\pm$ 1.12)
		0.3	3.26 ($\pm$ 0.12)	3.17 ($\pm$ 0.12)	1.38 ($\pm$ 0.62)	1.35 ($\pm$ 0.58)	13.41 ($\pm$ 1.94)	4.06 ($\pm$ 1.53)
		0.6	3.26 ($\pm$ 0.12)	2.84 ($\pm$ 0.10)	1.54 ($\pm$ 0.67)	1.46 ($\pm$ 0.65)	13.41 ($\pm$ 2.12)	8.36 ($\pm$ 1.28)
		0.9	3.26 ($\pm$ 0.11)	2.07 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.56 ($\pm$ 0.70)	13.45 ($\pm$ 2.32)	12.69 ($\pm$ 1.00)
$\mathbf{A}_3(t)$	1000	0.0	6.48 ($\pm$ 0.22)	6.59 ($\pm$ 0.23)	2.22 ($\pm$ 0.79)	2.21 ($\pm$ 0.75)	19.24 ($\pm$ 2.34)	2.95 ($\pm$ 2.18)
		0.3	6.51 ($\pm$ 0.18)	6.54 ($\pm$ 0.17)	2.43 ($\pm$ 1.08)	2.49 ($\pm$ 1.08)	19.03 ($\pm$ 2.81)	11.20 ($\pm$ 3.26)
		0.6	6.52 ($\pm$ 0.20)	6.28 ($\pm$ 0.17)	2.67 ($\pm$ 1.15)	2.69 ($\pm$ 1.15)	19.03 ($\pm$ 3.09)	22.52 ($\pm$ 2.99)
		0.9	6.53 ($\pm$ 0.22)	5.74 ($\pm$ 0.16)	2.91 ($\pm$ 1.19)	2.86 ($\pm$ 1.20)	19.06 ($\pm$ 3.45)	33.85 ($\pm$ 2.68)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.52 ($\pm$ 0.55)	1.47 ($\pm$ 0.56)	10.30 ($\pm$ 1.13)	1.39 ($\pm$ 1.04)
		0.3	5.05 ($\pm$ 0.12)	5.08 ($\pm$ 0.12)	1.41 ($\pm$ 0.58)	1.35 ($\pm$ 0.61)	10.84 ($\pm$ 1.29)	8.51 ($\pm$ 1.89)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.54 ($\pm$ 0.63)	1.46 ($\pm$ 0.66)	10.99 ($\pm$ 1.33)	16.96 ($\pm$ 1.81)
		0.9	5.06 ($\pm$ 0.13)	4.49 ($\pm$ 0.09)	1.67 ($\pm$ 0.68)	1.56 ($\pm$ 0.70)	11.22 ($\pm$ 1.37)	25.42 ($\pm$ 1.66)
$\mathbf{A}_4(t)$	1000	0.0	4.85 ($\pm$ 0.20)	4.86 ($\pm$ 0.20)	2.12 ($\pm$ 0.78)	2.05 ($\pm$ 0.71)	31.88 ($\pm$ 2.79)	2.95 ($\pm$ 2.01)
		0.3	4.82 ($\pm$ 0.18)	4.70 ($\pm$ 0.16)	2.39 ($\pm$ 1.03)	2.42 ($\pm$ 1.05)	31.72 ($\pm$ 3.45)	6.98 ($\pm$ 3.13)
		0.6	4.82 ($\pm$ 0.21)	4.26 ($\pm$ 0.15)	2.62 ($\pm$ 1.10)	2.61 ($\pm$ 1.12)	31.71 ($\pm$ 3.75)	14.11 ($\pm$ 2.52)
		0.9	4.82 ($\pm$ 0.24)	3.18 ($\pm$ 0.15)	2.85 ($\pm$ 1.14)	2.77 ($\pm$ 1.13)	31.69 ($\pm$ 4.10)	21.23 ($\pm$ 1.76)
	2000	0.0	3.00 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.25 ($\pm$ 0.47)	1.23 ($\pm$ 0.43)	15.24 ($\pm$ 1.94)	1.65 ($\pm$ 0.92)
		0.3	3.01 ($\pm$ 0.09)	2.89 ($\pm$ 0.09)	1.36 ($\pm$ 0.59)	1.31 ($\pm$ 0.57)	15.15 ($\pm$ 1.94)	3.71 ($\pm$ 1.47)
		0.6	3.02 ($\pm$ 0.09)	2.52 ($\pm$ 0.08)	1.51 ($\pm$ 0.65)	1.42 ($\pm$ 0.64)	15.14 ($\pm$ 2.11)	7.47 ($\pm$ 1.21)
		0.9	3.02 ($\pm$ 0.09)	1.62 ($\pm$ 0.06)	1.65 ($\pm$ 0.70)	1.51 ($\pm$ 0.68)	15.15 ($\pm$ 2.33)	11.25 ($\pm$ 0.85)
$\mathbf{A}_5(t)$	1000	0.0	7.85 ($\pm$ 0.23)	8.15 ($\pm$ 0.25)	2.57 ($\pm$ 1.07)	2.44 ($\pm$ 0.94)	26.70 ($\pm$ 2.27)	3.61 ($\pm$ 2.24)
		0.3	7.81 ($\pm$ 0.25)	8.05 ($\pm$ 0.25)	2.63 ($\pm$ 0.82)	2.60 ($\pm$ 0.94)	26.69 ($\pm$ 2.51)	14.59 ($\pm$ 3.55)
		0.6	7.81 ($\pm$ 0.26)	7.85 ($\pm$ 0.25)	2.85 ($\pm$ 0.86)	2.82 ($\pm$ 1.02)	26.63 ($\pm$ 2.58)	29.96 ($\pm$ 3.56)
		0.9	7.81 ($\pm$ 0.28)	7.45 ($\pm$ 0.27)	3.08 ($\pm$ 0.90)	3.02 ($\pm$ 1.08)	26.58 ($\pm$ 2.69)	45.32 ($\pm$ 3.51)
	2000	0.0	6.44 ($\pm$ 0.13)	6.74 ($\pm$ 0.14)	1.58 ($\pm$ 0.53)	1.48 ($\pm$ 0.47)	38.26 ($\pm$ 2.98)	2.13 ($\pm$ 1.50)
		0.3	6.46 ($\pm$ 0.13)	6.70 ($\pm$ 0.13)	1.67 ($\pm$ 0.63)	1.57 ($\pm$ 0.56)	37.88 ($\pm$ 2.37)	13.22 ($\pm$ 2.83)
		0.6	6.46 ($\pm$ 0.13)	6.54 ($\pm$ 0.13)	1.80 ($\pm$ 0.67)	1.69 ($\pm$ 0.60)	37.88 ($\pm$ 2.32)	26.55 ($\pm$ 2.68)
		0.9	6.46 ($\pm$ 0.13)	6.22 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.80 ($\pm$ 0.62)	37.90 ($\pm$ 2.35)	39.91 ($\pm$ 2.40)
$\mathbf{A}_6(t)$	1000	0.0	6.16 ($\pm$ 0.21)	6.23 ($\pm$ 0.22)	2.11 ($\pm$ 0.87)	2.11 ($\pm$ 0.79)	20.88 ($\pm$ 2.96)	3.10 ($\pm$ 2.08)
		0.3	6.10 ($\pm$ 0.21)	6.09 ($\pm$ 0.20)	2.62 ($\pm$ 1.05)	2.57 ($\pm$ 1.01)	20.76 ($\pm$ 3.59)	9.42 ($\pm$ 3.30)
		0.6	6.09 ($\pm$ 0.23)	5.78 ($\pm$ 0.20)	2.83 ($\pm$ 1.15)	2.75 ($\pm$ 1.09)	20.79 ($\pm$ 3.83)	19.70 ($\pm$ 2.82)
		0.9	6.08 ($\pm$ 0.24)	5.16 ($\pm$ 0.20)	3.04 ($\pm$ 1.23)	2.88 ($\pm$ 1.15)	20.80 ($\pm$ 4.15)	29.97 ($\pm$ 2.13)
	2000	0.0	4.58 ($\pm$ 0.10)	4.64 ($\pm$ 0.10)	1.16 ($\pm$ 0.45)	1.16 ($\pm$ 0.44)	6.98 ($\pm$ 1.29)	1.80 ($\pm$ 1.19)
		0.3	4.58 ($\pm$ 0.13)	4.59 ($\pm$ 0.12)	1.53 ($\pm$ 0.69)	1.52 ($\pm$ 0.64)	7.08 ($\pm$ 1.29)	7.02 ($\pm$ 1.88)
		0.6	4.59 ($\pm$ 0.13)	4.38 ($\pm$ 0.11)	1.68 ($\pm$ 0.75)	1.64 ($\pm$ 0.70)	7.21 ($\pm$ 1.49)	14.49 ($\pm$ 1.74)
		0.9	4.58 ($\pm$ 0.14)	3.94 ($\pm$ 0.08)	1.84 ($\pm$ 0.79)	1.75 ($\pm$ 0.74)	7.38 ($\pm$ 1.73)	22.00 ($\pm$ 1.61)

Table 3: MAE errors ($\times 10^{-2}$) for Case 1(c) - $\sigma_2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	4.28 ($\pm$ 0.14)	4.29 ($\pm$ 0.14)	1.82 ($\pm$ 0.68)	1.78 ($\pm$ 0.66)	20.98 ($\pm$ 2.10)	2.79 ($\pm$ 1.96)
		0.3	4.28 ($\pm$ 0.18)	4.19 ($\pm$ 0.16)	2.08 ($\pm$ 0.85)	2.12 ($\pm$ 0.89)	20.91 ($\pm$ 2.62)	7.39 ($\pm$ 3.17)
		0.6	4.28 ($\pm$ 0.20)	3.84 ($\pm$ 0.15)	2.27 ($\pm$ 0.92)	2.29 ($\pm$ 0.94)	20.89 ($\pm$ 2.80)	14.77 ($\pm$ 2.63)
		0.9	4.28 ($\pm$ 0.23)	3.06 ($\pm$ 0.15)	2.45 ($\pm$ 0.96)	2.40 ($\pm$ 0.96)	20.85 ($\pm$ 3.03)	22.15 ($\pm$ 1.96)
	2000	0.0	2.85 ($\pm$ 0.09)	2.86 ($\pm$ 0.09)	1.09 ($\pm$ 0.38)	1.04 ($\pm$ 0.38)	8.56 ($\pm$ 1.44)	1.51 ($\pm$ 0.91)
		0.3	2.87 ($\pm$ 0.09)	2.79 ($\pm$ 0.09)	1.23 ($\pm$ 0.52)	1.13 ($\pm$ 0.51)	8.47 ($\pm$ 1.43)	4.17 ($\pm$ 1.73)
		0.6	2.88 ($\pm$ 0.09)	2.53 ($\pm$ 0.09)	1.35 ($\pm$ 0.57)	1.23 ($\pm$ 0.56)	8.41 ($\pm$ 1.58)	8.51 ($\pm$ 1.42)
		0.9	2.88 ($\pm$ 0.10)	1.96 ($\pm$ 0.06)	1.48 ($\pm$ 0.61)	1.31 ($\pm$ 0.60)	8.37 ($\pm$ 1.77)	12.91 ($\pm$ 1.01)
$\mathbf{A}_2(t)$	1000	0.0	4.29 ($\pm$ 0.17)	4.31 ($\pm$ 0.17)	1.83 ($\pm$ 0.69)	1.77 ($\pm$ 0.67)	21.00 ($\pm$ 2.15)	3.03 ($\pm$ 2.04)
		0.3	4.24 ($\pm$ 0.15)	4.16 ($\pm$ 0.14)	2.08 ($\pm$ 0.83)	2.13 ($\pm$ 0.87)	20.97 ($\pm$ 2.70)	6.68 ($\pm$ 2.98)
		0.6	4.24 ($\pm$ 0.17)	3.81 ($\pm$ 0.13)	2.27 ($\pm$ 0.89)	2.30 ($\pm$ 0.93)	20.95 ($\pm$ 2.87)	14.04 ($\pm$ 2.36)
		0.9	4.23 ($\pm$ 0.18)	3.00 ($\pm$ 0.13)	2.48 ($\pm$ 0.92)	2.41 ($\pm$ 0.97)	20.91 ($\pm$ 3.09)	21.43 ($\pm$ 1.49)
	2000	0.0	2.83 ($\pm$ 0.07)	2.84 ($\pm$ 0.07)	1.04 ($\pm$ 0.40)	1.01 ($\pm$ 0.39)	8.68 ($\pm$ 1.45)	1.75 ($\pm$ 1.15)
		0.3	2.83 ($\pm$ 0.10)	2.76 ($\pm$ 0.10)	1.22 ($\pm$ 0.53)	1.15 ($\pm$ 0.51)	8.56 ($\pm$ 1.41)	3.87 ($\pm$ 1.72)
		0.6	2.83 ($\pm$ 0.10)	2.49 ($\pm$ 0.08)	1.36 ($\pm$ 0.56)	1.25 ($\pm$ 0.58)	8.49 ($\pm$ 1.58)	8.19 ($\pm$ 1.38)
		0.9	2.83 ($\pm$ 0.10)	1.87 ($\pm$ 0.06)	1.49 ($\pm$ 0.59)	1.34 ($\pm$ 0.61)	8.42 ($\pm$ 1.81)	12.56 ($\pm$ 0.99)
$\mathbf{A}_3(t)$	1000	0.0	5.84 ($\pm$ 0.19)	5.96 ($\pm$ 0.20)	1.94 ($\pm$ 0.68)	1.95 ($\pm$ 0.68)	9.43 ($\pm$ 1.33)	3.00 ($\pm$ 2.14)
		0.3	5.88 ($\pm$ 0.15)	5.94 ($\pm$ 0.15)	2.12 ($\pm$ 0.98)	2.15 ($\pm$ 0.90)	9.57 ($\pm$ 1.61)	11.74 ($\pm$ 3.09)
		0.6	5.88 ($\pm$ 0.16)	5.74 ($\pm$ 0.14)	2.32 ($\pm$ 1.05)	2.32 ($\pm$ 0.95)	9.61 ($\pm$ 1.60)	23.73 ($\pm$ 2.84)
		0.9	5.89 ($\pm$ 0.17)	5.34 ($\pm$ 0.13)	2.54 ($\pm$ 1.06)	2.45 ($\pm$ 1.00)	9.69 ($\pm$ 1.68)	35.74 ($\pm$ 2.62)
	2000	0.0	4.72 ($\pm$ 0.10)	4.84 ($\pm$ 0.11)	1.36 ($\pm$ 0.51)	1.29 ($\pm$ 0.51)	10.46 ($\pm$ 0.95)	1.38 ($\pm$ 1.05)
		0.3	4.72 ($\pm$ 0.11)	4.80 ($\pm$ 0.10)	1.28 ($\pm$ 0.50)	1.17 ($\pm$ 0.52)	10.69 ($\pm$ 1.10)	9.40 ($\pm$ 2.06)
		0.6	4.73 ($\pm$ 0.12)	4.65 ($\pm$ 0.10)	1.39 ($\pm$ 0.55)	1.26 ($\pm$ 0.56)	10.78 ($\pm$ 1.15)	18.86 ($\pm$ 1.97)
		0.9	4.73 ($\pm$ 0.12)	4.37 ($\pm$ 0.08)	1.50 ($\pm$ 0.59)	1.34 ($\pm$ 0.60)	10.92 ($\pm$ 1.25)	28.35 ($\pm$ 1.83)
$\mathbf{A}_4(t)$	1000	0.0	4.07 ($\pm$ 0.15)	4.07 ($\pm$ 0.16)	1.81 ($\pm$ 0.67)	1.79 ($\pm$ 0.64)	22.43 ($\pm$ 2.13)	2.89 ($\pm$ 1.95)
		0.3	4.05 ($\pm$ 0.15)	3.95 ($\pm$ 0.14)	2.07 ($\pm$ 0.86)	2.09 ($\pm$ 0.89)	22.37 ($\pm$ 2.61)	6.57 ($\pm$ 3.04)
		0.6	4.06 ($\pm$ 0.18)	3.57 ($\pm$ 0.13)	2.26 ($\pm$ 0.93)	2.27 ($\pm$ 0.94)	22.34 ($\pm$ 2.78)	13.34 ($\pm$ 2.41)
		0.9	4.05 ($\pm$ 0.20)	2.65 ($\pm$ 0.13)	2.46 ($\pm$ 0.96)	2.38 ($\pm$ 0.96)	22.31 ($\pm$ 3.01)	20.13 ($\pm$ 1.62)
	2000	0.0	2.55 ($\pm$ 0.08)	2.55 ($\pm$ 0.08)	1.10 ($\pm$ 0.40)	1.05 ($\pm$ 0.38)	10.37 ($\pm$ 1.44)	1.56 ($\pm$ 0.90)
		0.3	2.56 ($\pm$ 0.08)	2.45 ($\pm$ 0.08)	1.19 ($\pm$ 0.50)	1.11 ($\pm$ 0.50)	10.27 ($\pm$ 1.46)	3.45 ($\pm$ 1.54)
		0.6	2.56 ($\pm$ 0.08)	2.15 ($\pm$ 0.07)	1.33 ($\pm$ 0.55)	1.21 ($\pm$ 0.55)	10.21 ($\pm$ 1.61)	7.01 ($\pm$ 1.30)
		0.9	2.56 ($\pm$ 0.08)	1.39 ($\pm$ 0.05)	1.45 ($\pm$ 0.59)	1.28 ($\pm$ 0.59)	10.14 ($\pm$ 1.82)	10.67 ($\pm$ 0.83)
$\mathbf{A}_5(t)$	1000	0.0	7.99 ($\pm$ 0.23)	8.31 ($\pm$ 0.25)	2.56 ($\pm$ 1.06)	2.50 ($\pm$ 1.00)	32.41 ($\pm$ 2.33)	3.50 ($\pm$ 2.03)
		0.3	7.96 ($\pm$ 0.26)	8.21 ($\pm$ 0.26)	2.64 ($\pm$ 0.83)	2.69 ($\pm$ 0.96)	32.40 ($\pm$ 2.47)	14.82 ($\pm$ 3.62)
		0.6	7.96 ($\pm$ 0.28)	8.00 ($\pm$ 0.27)	2.87 ($\pm$ 0.90)	2.92 ($\pm$ 1.03)	32.35 ($\pm$ 2.51)	30.51 ($\pm$ 3.66)
		0.9	7.97 ($\pm$ 0.30)	7.60 ($\pm$ 0.29)	3.09 ($\pm$ 0.96)	3.12 ($\pm$ 1.09)	32.35 ($\pm$ 2.59)	46.21 ($\pm$ 3.67)
	2000	0.0	6.52 ($\pm$ 0.13)	6.83 ($\pm$ 0.14)	1.58 ($\pm$ 0.56)	1.53 ($\pm$ 0.49)	40.76 ($\pm$ 2.52)	1.90 ($\pm$ 1.35)
		0.3	6.55 ($\pm$ 0.13)	6.80 ($\pm$ 0.13)	1.67 ($\pm$ 0.62)	1.61 ($\pm$ 0.58)	40.52 ($\pm$ 1.85)	13.24 ($\pm$ 2.74)
		0.6	6.55 ($\pm$ 0.13)	6.63 ($\pm$ 0.13)	1.80 ($\pm$ 0.65)	1.74 ($\pm$ 0.62)	40.56 ($\pm$ 1.81)	26.77 ($\pm$ 2.60)
		0.9	6.55 ($\pm$ 0.14)	6.29 ($\pm$ 0.13)	1.93 ($\pm$ 0.69)	1.85 ($\pm$ 0.64)	40.62 ($\pm$ 1.89)	40.36 ($\pm$ 2.34)
$\mathbf{A}_6(t)$	1000	0.0	6.26 ($\pm$ 0.22)	6.34 ($\pm$ 0.22)	2.08 ($\pm$ 0.86)	2.14 ($\pm$ 0.78)	22.13 ($\pm$ 2.88)	3.02 ($\pm$ 2.21)
		0.3	6.20 ($\pm$ 0.21)	6.19 ($\pm$ 0.20)	2.64 ($\pm$ 1.05)	2.55 ($\pm$ 1.01)	21.97 ($\pm$ 3.67)	9.59 ($\pm$ 3.29)
		0.6	6.19 ($\pm$ 0.23)	5.88 ($\pm$ 0.20)	2.86 ($\pm$ 1.15)	2.73 ($\pm$ 1.10)	21.94 ($\pm$ 3.87)	20.22 ($\pm$ 2.83)
		0.9	6.18 ($\pm$ 0.26)	5.26 ($\pm$ 0.21)	3.07 ($\pm$ 1.24)	2.88 ($\pm$ 1.14)	21.92 ($\pm$ 4.10)	30.87 ($\pm$ 2.19)
	2000	0.0	4.61 ($\pm$ 0.10)	4.68 ($\pm$ 0.10)	1.18 ($\pm$ 0.46)	1.15 ($\pm$ 0.44)	7.91 ($\pm$ 1.13)	1.58 ($\pm$ 1.01)
		0.3	4.64 ($\pm$ 0.14)	4.64 ($\pm$ 0.12)	1.54 ($\pm$ 0.64)	1.50 ($\pm$ 0.68)	8.20 ($\pm$ 1.19)	7.03 ($\pm$ 1.92)
		0.6	4.64 ($\pm$ 0.14)	4.42 ($\pm$ 0.11)	1.69 ($\pm$ 0.69)	1.62 ($\pm$ 0.74)	8.24 ($\pm$ 1.26)	14.72 ($\pm$ 1.77)
		0.9	4.63 ($\pm$ 0.14)	3.97 ($\pm$ 0.09)	1.84 ($\pm$ 0.73)	1.74 ($\pm$ 0.78)	8.31 ($\pm$ 1.40)	22.50 ($\pm$ 1.60)

Table 4: MAE errors ($\times 10^{-2}$) for Case 1(d) - $\sigma_3(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	1000	0.0	5.17 ($\pm$ 0.18)	5.18 ($\pm$ 0.19)	2.08 ($\pm$ 0.79)	2.09 ($\pm$ 0.75)	31.65 ($\pm$ 2.68)	2.92 ($\pm$ 2.04)
		0.3	5.15 ($\pm$ 0.21)	5.03 ($\pm$ 0.18)	2.45 ($\pm$ 1.03)	2.44 ($\pm$ 1.05)	31.52 ($\pm$ 3.55)	7.70 ($\pm$ 3.10)
		0.6	5.16 ($\pm$ 0.24)	4.62 ($\pm$ 0.17)	2.68 ($\pm$ 1.10)	2.64 ($\pm$ 1.11)	31.47 ($\pm$ 3.79)	15.65 ($\pm$ 2.60)
		0.9	5.16 ($\pm$ 0.28)	3.67 ($\pm$ 0.19)	2.90 ($\pm$ 1.16)	2.79 ($\pm$ 1.12)	31.40 ($\pm$ 4.05)	23.60 ($\pm$ 2.02)
	2000	0.0	3.35 ($\pm$ 0.10)	3.35 ($\pm$ 0.10)	1.25 ($\pm$ 0.43)	1.23 ($\pm$ 0.43)	14.39 ($\pm$ 1.76)	1.31 ($\pm$ 0.81)
		0.3	3.38 ($\pm$ 0.11)	3.27 ($\pm$ 0.11)	1.38 ($\pm$ 0.59)	1.35 ($\pm$ 0.60)	14.46 ($\pm$ 1.71)	4.27 ($\pm$ 1.69)
		0.6	3.39 ($\pm$ 0.11)	2.94 ($\pm$ 0.10)	1.53 ($\pm$ 0.64)	1.46 ($\pm$ 0.66)	14.38 ($\pm$ 1.97)	8.81 ($\pm$ 1.48)
		0.9	3.39 ($\pm$ 0.11)	2.19 ($\pm$ 0.07)	1.67 ($\pm$ 0.69)	1.55 ($\pm$ 0.70)	14.32 ($\pm$ 2.29)	13.44 ($\pm$ 1.10)
$\mathbf{A}_2(t)$	1000	0.0	5.19 ($\pm$ 0.22)	5.20 ($\pm$ 0.22)	2.08 ($\pm$ 0.80)	2.11 ($\pm$ 0.76)	31.67 ($\pm$ 2.75)	3.07 ($\pm$ 2.20)
		0.3	5.13 ($\pm$ 0.18)	5.02 ($\pm$ 0.17)	2.46 ($\pm$ 1.00)	2.45 ($\pm$ 1.01)	31.56 ($\pm$ 3.61)	7.11 ($\pm$ 3.01)
		0.6	5.13 ($\pm$ 0.20)	4.60 ($\pm$ 0.16)	2.69 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	31.52 ($\pm$ 3.83)	15.04 ($\pm$ 2.38)
		0.9	5.13 ($\pm$ 0.23)	3.62 ($\pm$ 0.16)	2.92 ($\pm$ 1.16)	2.80 ($\pm$ 1.11)	31.47 ($\pm$ 4.09)	23.00 ($\pm$ 1.58)
	2000	0.0	3.34 ($\pm$ 0.08)	3.33 ($\pm$ 0.08)	1.20 ($\pm$ 0.46)	1.19 ($\pm$ 0.45)	14.82 ($\pm$ 1.71)	1.45 ($\pm$ 0.96)
		0.3	3.36 ($\pm$ 0.12)	3.25 ($\pm$ 0.12)	1.39 ($\pm$ 0.59)	1.35 ($\pm$ 0.63)	14.94 ($\pm$ 1.66)	4.05 ($\pm$ 1.61)
		0.6	3.36 ($\pm$ 0.12)	2.91 ($\pm$ 0.11)	1.54 ($\pm$ 0.64)	1.47 ($\pm$ 0.70)	14.87 ($\pm$ 1.93)	8.58 ($\pm$ 1.36)
		0.9	3.36 ($\pm$ 0.12)	2.12 ($\pm$ 0.08)	1.69 ($\pm$ 0.69)	1.58 ($\pm$ 0.74)	14.79 ($\pm$ 2.25)	13.19 ($\pm$ 1.04)
$\mathbf{A}_3(t)$	1000	0.0	6.60 ($\pm$ 0.23)	6.71 ($\pm$ 0.24)	2.21 ($\pm$ 0.77)	2.23 ($\pm$ 0.74)	19.12 ($\pm$ 2.45)	2.91 ($\pm$ 2.24)
		0.3	6.62 ($\pm$ 0.19)	6.66 ($\pm$ 0.17)	2.48 ($\pm$ 1.09)	2.49 ($\pm$ 1.09)	19.06 ($\pm$ 3.18)	11.24 ($\pm$ 3.14)
		0.6	6.63 ($\pm$ 0.20)	6.39 ($\pm$ 0.16)	2.72 ($\pm$ 1.16)	2.69 ($\pm$ 1.17)	19.13 ($\pm$ 3.23)	22.90 ($\pm$ 2.84)
		0.9	6.64 ($\pm$ 0.22)	5.86 ($\pm$ 0.16)	2.96 ($\pm$ 1.20)	2.85 ($\pm$ 1.23)	19.23 ($\pm$ 3.26)	34.59 ($\pm$ 2.55)
	2000	0.0	5.10 ($\pm$ 0.11)	5.19 ($\pm$ 0.12)	1.48 ($\pm$ 0.55)	1.47 ($\pm$ 0.57)	12.79 ($\pm$ 1.33)	1.30 ($\pm$ 0.97)
		0.3	5.11 ($\pm$ 0.12)	5.14 ($\pm$ 0.12)	1.40 ($\pm$ 0.60)	1.37 ($\pm$ 0.60)	13.23 ($\pm$ 1.41)	8.47 ($\pm$ 1.87)
		0.6	5.12 ($\pm$ 0.13)	4.94 ($\pm$ 0.11)	1.53 ($\pm$ 0.66)	1.48 ($\pm$ 0.64)	13.33 ($\pm$ 1.53)	17.14 ($\pm$ 1.83)
		0.9	5.12 ($\pm$ 0.13)	4.53 ($\pm$ 0.09)	1.65 ($\pm$ 0.72)	1.58 ($\pm$ 0.68)	13.45 ($\pm$ 1.70)	25.86 ($\pm$ 1.70)
$\mathbf{A}_4(t)$	1000	0.0	4.97 ($\pm$ 0.21)	4.98 ($\pm$ 0.21)	2.08 ($\pm$ 0.77)	2.10 ($\pm$ 0.72)	33.11 ($\pm$ 2.70)	2.93 ($\pm$ 2.14)
		0.3	4.94 ($\pm$ 0.19)	4.82 ($\pm$ 0.16)	2.43 ($\pm$ 1.03)	2.42 ($\pm$ 1.04)	32.96 ($\pm$ 3.54)	7.05 ($\pm$ 3.01)
		0.6	4.95 ($\pm$ 0.21)	4.37 ($\pm$ 0.15)	2.67 ($\pm$ 1.11)	2.61 ($\pm$ 1.13)	32.92 ($\pm$ 3.77)	14.53 ($\pm$ 2.42)
		0.9	4.95 ($\pm$ 0.25)	3.30 ($\pm$ 0.16)	2.90 ($\pm$ 1.16)	2.77 ($\pm$ 1.13)	32.86 ($\pm$ 4.03)	22.04 ($\pm$ 1.74)
	2000	0.0	3.06 ($\pm$ 0.09)	3.06 ($\pm$ 0.09)	1.26 ($\pm$ 0.44)	1.24 ($\pm$ 0.45)	16.05 ($\pm$ 1.84)	1.38 ($\pm$ 0.82)
		0.3	3.08 ($\pm$ 0.10)	2.96 ($\pm$ 0.10)	1.35 ($\pm$ 0.58)	1.32 ($\pm$ 0.59)	16.08 ($\pm$ 1.78)	3.71 ($\pm$ 1.48)
		0.6	3.08 ($\pm$ 0.09)	2.59 ($\pm$ 0.08)	1.50 ($\pm$ 0.64)	1.43 ($\pm$ 0.65)	15.99 ($\pm$ 2.05)	7.66 ($\pm$ 1.36)
		0.9	3.09 ($\pm$ 0.09)	1.66 ($\pm$ 0.06)	1.64 ($\pm$ 0.69)	1.53 ($\pm$ 0.69)	15.89 ($\pm$ 2.39)	11.73 ($\pm$ 0.95)
$\mathbf{A}_5(t)$	1000	0.0	7.26 ($\pm$ 0.19)	7.59 ($\pm$ 0.22)	2.34 ($\pm$ 0.93)	2.14 ($\pm$ 0.85)	25.26 ($\pm$ 2.29)	3.63 ($\pm$ 2.11)
		0.3	7.23 ($\pm$ 0.23)	7.52 ($\pm$ 0.24)	2.31 ($\pm$ 0.75)	2.33 ($\pm$ 0.83)	25.11 ($\pm$ 2.60)	15.73 ($\pm$ 3.80)
		0.6	7.23 ($\pm$ 0.25)	7.36 ($\pm$ 0.25)	2.51 ($\pm$ 0.78)	2.51 ($\pm$ 0.88)	25.06 ($\pm$ 2.68)	32.31 ($\pm$ 3.85)
		0.9	7.23 ($\pm$ 0.26)	7.08 ($\pm$ 0.26)	2.69 ($\pm$ 0.80)	2.65 ($\pm$ 0.96)	25.03 ($\pm$ 2.79)	48.89 ($\pm$ 3.85)
	2000	0.0	6.14 ($\pm$ 0.13)	6.49 ($\pm$ 0.14)	1.47 ($\pm$ 0.49)	1.33 ($\pm$ 0.44)	39.37 ($\pm$ 2.68)	2.28 ($\pm$ 1.74)
		0.3	6.16 ($\pm$ 0.12)	6.46 ($\pm$ 0.12)	1.52 ($\pm$ 0.58)	1.39 ($\pm$ 0.45)	38.85 ($\pm$ 2.12)	14.64 ($\pm$ 2.96)
		0.6	6.16 ($\pm$ 0.12)	6.34 ($\pm$ 0.12)	1.63 ($\pm$ 0.62)	1.49 ($\pm$ 0.48)	38.90 ($\pm$ 2.11)	29.59 ($\pm$ 2.81)
		0.9	6.15 ($\pm$ 0.13)	6.12 ($\pm$ 0.13)	1.74 ($\pm$ 0.66)	1.58 ($\pm$ 0.49)	39.00 ($\pm$ 2.17)	44.58 ($\pm$ 2.55)
$\mathbf{A}_6(t)$	1000	0.0	5.57 ($\pm$ 0.18)	5.63 ($\pm$ 0.18)	1.88 ($\pm$ 0.72)	1.85 ($\pm$ 0.69)	12.29 ($\pm$ 1.89)	2.83 ($\pm$ 2.15)
		0.3	5.52 ($\pm$ 0.18)	5.52 ($\pm$ 0.18)	2.27 ($\pm$ 0.86)	2.32 ($\pm$ 0.90)	12.39 ($\pm$ 2.44)	9.89 ($\pm$ 3.12)
		0.6	5.51 ($\pm$ 0.19)	5.28 ($\pm$ 0.18)	2.45 ($\pm$ 0.94)	2.47 ($\pm$ 0.98)	12.43 ($\pm$ 2.53)	20.64 ($\pm$ 2.74)
		0.9	5.49 ($\pm$ 0.21)	4.80 ($\pm$ 0.18)	2.63 ($\pm$ 0.98)	2.58 ($\pm$ 1.03)	12.48 ($\pm$ 2.66)	31.40 ($\pm$ 2.19)
	2000	0.0	4.29 ($\pm$ 0.09)	4.35 ($\pm$ 0.10)	1.04 ($\pm$ 0.40)	1.03 ($\pm$ 0.38)	9.57 ($\pm$ 1.10)	1.83 ($\pm$ 1.18)
		0.3	4.30 ($\pm$ 0.11)	4.32 ($\pm$ 0.10)	1.39 ($\pm$ 0.58)	1.36 ($\pm$ 0.56)	9.76 ($\pm$ 1.20)	7.66 ($\pm$ 1.92)
		0.6	4.30 ($\pm$ 0.11)	4.16 ($\pm$ 0.09)	1.52 ($\pm$ 0.63)	1.47 ($\pm$ 0.61)	9.77 ($\pm$ 1.36)	15.93 ($\pm$ 1.74)
		0.9	4.29 ($\pm$ 0.11)	3.83 ($\pm$ 0.07)	1.65 ($\pm$ 0.67)	1.58 ($\pm$ 0.65)	9.79 ($\pm$ 1.54)	24.27 ($\pm$ 1.57)

Below we present three examples, with fitted-versus-true curves for $\mathbf{A}_2$, $\mathbf{A}_4$, and $\mathbf{A}_6$, respectively. These plots show that our method is effective across a range of different $\mathbf{A}(t)$ configurations.

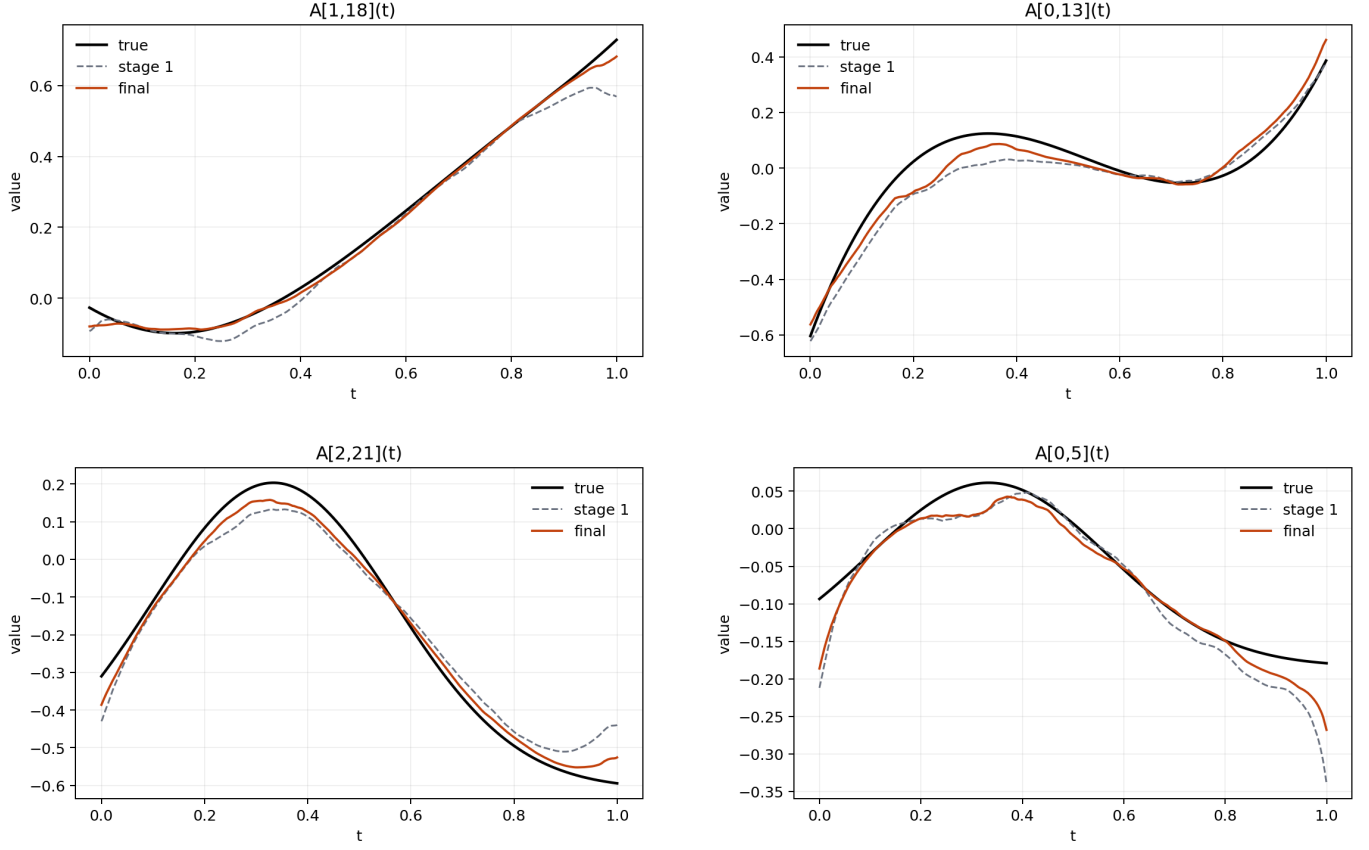


Figure 1: Selected estimated coefficient-function curves from Case 1.

We also present 3 examples of fitted-versus-true curves for $\sigma^2(t)$. They show that our method is effective for estimating the variance function across different settings.

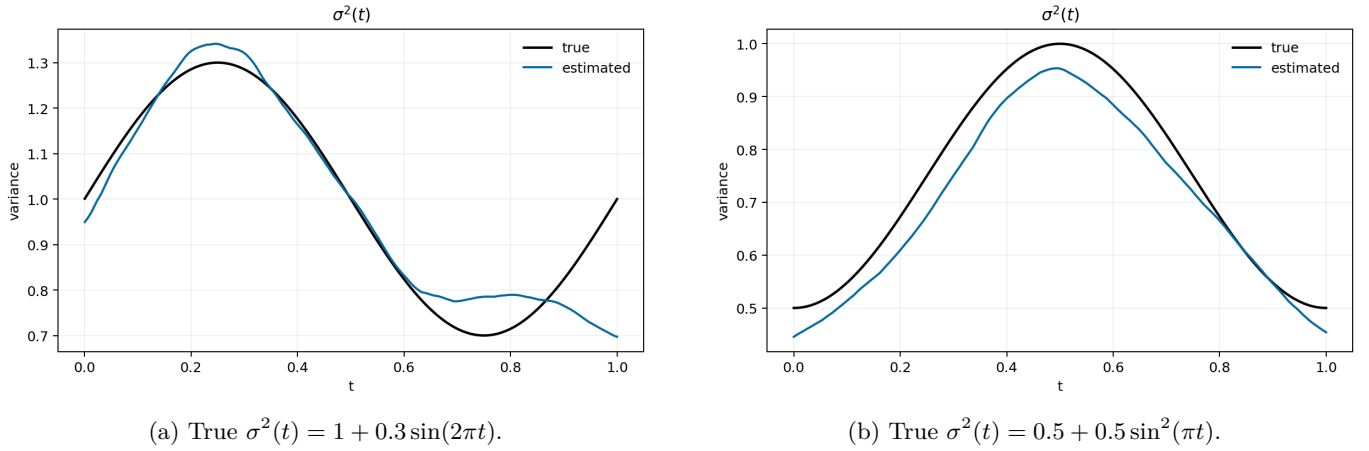


Figure 2: Selected estimated variance-function curves from Case 1.

Case 2: 3D Tensor Covariate

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 48$, and split by $p'_1 = p'_2 = p'_3 = 16$ such that by $S^{(1)} = S^{(2)} = S^{(3)} = 3$. Set the rank $R = 6$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\begin{aligned} \alpha_{1rs}(t) &= 5\sqrt{\frac{rs}{RS}}(t-0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}}(e^{-(3t-1)^2} - 0.75), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}}\sin(2\pi(t-0.5)), \\ \alpha_{4rs}(t) &= \sqrt{\frac{rs}{RS}}[0.45 \cdot 5(t-0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20\sin(2\pi(t-0.5))], \\ \alpha_{5rs}(t) &= \sqrt{\frac{rs}{RS}}\left[1.10\exp\left\{-\frac{(t-0.30)^2}{2(0.08)^2}\right\} - 0.95\exp\left\{-\frac{(t-0.72)^2}{2(0.11)^2}\right\}\right], \\ \alpha_{6rs}(t) &= \sqrt{\frac{rs}{RS}}[18(t-0.2)(t-0.55)(t-0.85) - 0.05] \end{aligned}$$

Representative coefficient functions: $R = 6, S = 27, r = 3, s = 14$

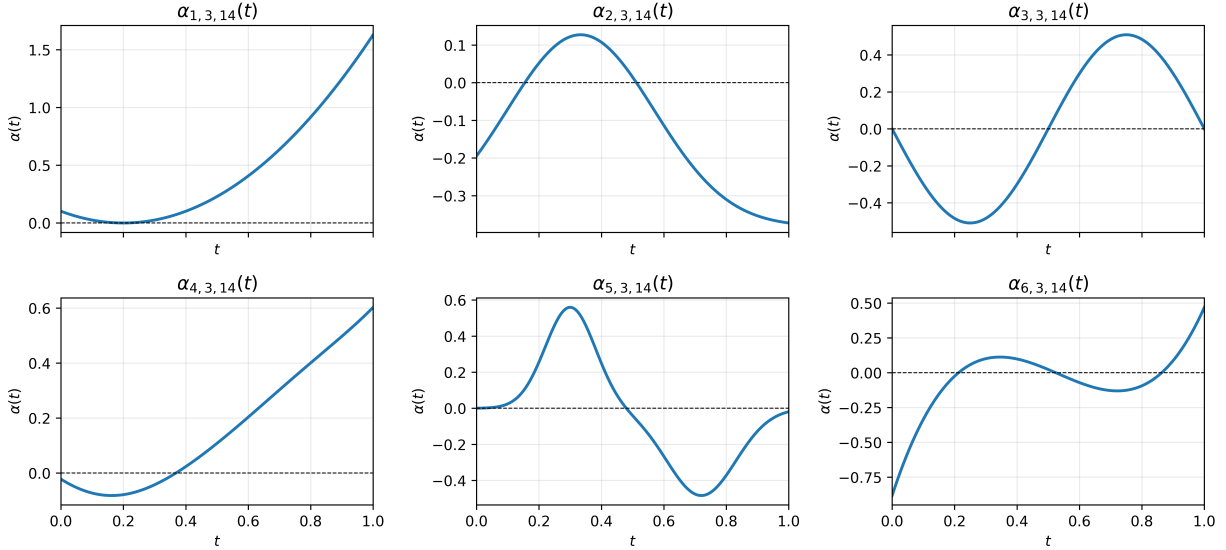


Figure 3: Representative coefficient functions in Case 2.

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. We will vary $\sigma^2(t)$ to 4 subcases as Case 1(a)-(d).

$$\sigma_0^2(t) \equiv 1, \quad \sigma_1^2(t) = 1 + 0.3\sin(2\pi t), \quad \sigma_2^2(t) = 1 + 0.25\cos(2\pi t) + 0.1\sin(4\pi t), \quad \text{and} \quad \sigma_3^2(t) = 0.5 + 0.5\sin^2(\pi t). \quad (3)$$

Simulation Results. We vary $n = 2000/5000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{6rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Table 5: MAE errors ($\times 10^{-2}$) for Case 2(a) - $\sigma_0^2(t) \equiv 1$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.51 ($\pm$ 0.10)	3.53 ($\pm$ 0.10)	1.55 ($\pm$ 0.65)	1.53 ($\pm$ 0.66)	18.86 ($\pm$ 2.34)	1.96 ($\pm$ 1.45)
		0.3	3.51 ($\pm$ 0.10)	3.44 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.69 ($\pm$ 0.71)	18.64 ($\pm$ 2.18)	6.54 ($\pm$ 2.61)
		0.6	3.51 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.86 ($\pm$ 0.77)	1.83 ($\pm$ 0.75)	18.61 ($\pm$ 2.62)	13.45 ($\pm$ 2.06)
		0.9	3.52 ($\pm$ 0.11)	2.58 ($\pm$ 0.08)	2.02 ($\pm$ 0.83)	1.96 ($\pm$ 0.77)	18.66 ($\pm$ 2.97)	20.39 ($\pm$ 1.34)
	5000	0.0	2.42 ($\pm$ 0.06)	2.43 ($\pm$ 0.06)	0.90 ($\pm$ 0.36)	0.91 ($\pm$ 0.37)	4.58 ($\pm$ 0.92)	1.15 ($\pm$ 0.78)
		0.3	2.39 ($\pm$ 0.06)	2.35 ($\pm$ 0.06)	0.95 ($\pm$ 0.30)	0.94 ($\pm$ 0.30)	4.61 ($\pm$ 0.98)	3.99 ($\pm$ 1.29)
		0.6	2.39 ($\pm$ 0.06)	2.20 ($\pm$ 0.06)	1.04 ($\pm$ 0.34)	1.03 ($\pm$ 0.32)	4.78 ($\pm$ 1.16)	7.73 ($\pm$ 1.04)
		0.9	2.39 ($\pm$ 0.06)	1.89 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.11 ($\pm$ 0.34)	5.03 ($\pm$ 1.35)	11.42 ($\pm$ 0.64)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.52 ($\pm$ 0.72)	1.50 ($\pm$ 0.73)	19.09 ($\pm$ 2.34)	1.68 ($\pm$ 1.37)
		0.3	3.45 ($\pm$ 0.09)	3.39 ($\pm$ 0.10)	1.75 ($\pm$ 0.68)	1.73 ($\pm$ 0.67)	18.90 ($\pm$ 2.18)	6.28 ($\pm$ 2.39)
		0.6	3.45 ($\pm$ 0.09)	3.11 ($\pm$ 0.09)	1.91 ($\pm$ 0.74)	1.87 ($\pm$ 0.74)	18.89 ($\pm$ 2.55)	13.04 ($\pm$ 2.10)
		0.9	3.45 ($\pm$ 0.10)	2.47 ($\pm$ 0.08)	2.06 ($\pm$ 0.80)	1.99 ($\pm$ 0.78)	18.92 ($\pm$ 2.91)	19.93 ($\pm$ 1.51)
	5000	0.0	2.35 ($\pm$ 0.05)	2.36 ($\pm$ 0.05)	0.91 ($\pm$ 0.38)	0.91 ($\pm$ 0.39)	6.60 ($\pm$ 0.93)	1.21 ($\pm$ 0.73)
		0.3	2.33 ($\pm$ 0.05)	2.29 ($\pm$ 0.05)	0.92 ($\pm$ 0.30)	0.91 ($\pm$ 0.29)	6.61 ($\pm$ 0.96)	3.91 ($\pm$ 1.23)
		0.6	2.32 ($\pm$ 0.05)	2.12 ($\pm$ 0.04)	1.02 ($\pm$ 0.32)	1.00 ($\pm$ 0.31)	6.73 ($\pm$ 1.09)	7.63 ($\pm$ 0.99)
		0.9	2.32 ($\pm$ 0.05)	1.74 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.33)	6.88 ($\pm$ 1.27)	11.32 ($\pm$ 0.66)
$\mathbf{A}_3(t)$	2000	0.0	5.71 ($\pm$ 0.10)	5.91 ($\pm$ 0.10)	1.69 ($\pm$ 0.66)	1.63 ($\pm$ 0.65)	19.93 ($\pm$ 2.11)	1.87 ($\pm$ 1.71)
		0.3	5.71 ($\pm$ 0.11)	5.87 ($\pm$ 0.11)	1.87 ($\pm$ 0.86)	1.84 ($\pm$ 0.89)	20.13 ($\pm$ 2.11)	13.90 ($\pm$ 2.51)
		0.6	5.72 ($\pm$ 0.11)	5.73 ($\pm$ 0.11)	2.02 ($\pm$ 0.93)	1.97 ($\pm$ 0.96)	20.22 ($\pm$ 2.45)	27.57 ($\pm$ 2.09)
		0.9	5.73 ($\pm$ 0.11)	5.47 ($\pm$ 0.10)	2.16 ($\pm$ 0.99)	2.08 ($\pm$ 1.02)	20.31 ($\pm$ 2.82)	41.25 ($\pm$ 1.62)
	5000	0.0	4.90 ($\pm$ 0.06)	5.11 ($\pm$ 0.06)	1.03 ($\pm$ 0.36)	1.03 ($\pm$ 0.37)	36.89 ($\pm$ 1.55)	1.02 ($\pm$ 0.75)
		0.3	4.90 ($\pm$ 0.07)	5.08 ($\pm$ 0.07)	1.04 ($\pm$ 0.44)	1.04 ($\pm$ 0.43)	36.94 ($\pm$ 1.76)	11.98 ($\pm$ 1.32)
		0.6	4.90 ($\pm$ 0.07)	5.00 ($\pm$ 0.06)	1.13 ($\pm$ 0.46)	1.13 ($\pm$ 0.46)	36.85 ($\pm$ 1.89)	23.54 ($\pm$ 1.24)
		0.9	4.91 ($\pm$ 0.06)	4.84 ($\pm$ 0.05)	1.21 ($\pm$ 0.49)	1.21 ($\pm$ 0.48)	36.75 ($\pm$ 2.08)	35.04 ($\pm$ 1.14)
$\mathbf{A}_4(t)$	2000	0.0	3.13 ($\pm$ 0.09)	3.14 ($\pm$ 0.09)	1.51 ($\pm$ 0.67)	1.49 ($\pm$ 0.67)	22.56 ($\pm$ 2.39)	1.83 ($\pm$ 1.46)
		0.3	3.12 ($\pm$ 0.08)	3.03 ($\pm$ 0.08)	1.68 ($\pm$ 0.70)	1.67 ($\pm$ 0.71)	22.38 ($\pm$ 2.22)	5.44 ($\pm$ 2.37)
		0.6	3.13 ($\pm$ 0.09)	2.71 ($\pm$ 0.07)	1.85 ($\pm$ 0.77)	1.81 ($\pm$ 0.77)	22.33 ($\pm$ 2.70)	11.14 ($\pm$ 1.88)
		0.9	3.13 ($\pm$ 0.10)	1.91 ($\pm$ 0.06)	2.00 ($\pm$ 0.82)	1.94 ($\pm$ 0.80)	22.29 ($\pm$ 3.24)	16.95 ($\pm$ 1.02)
	5000	0.0	1.87 ($\pm$ 0.04)	1.87 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.91 ($\pm$ 0.34)	8.40 ($\pm$ 1.33)	1.13 ($\pm$ 0.74)
		0.3	1.84 ($\pm$ 0.05)	1.78 ($\pm$ 0.05)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	8.34 ($\pm$ 1.30)	2.81 ($\pm$ 1.20)
		0.6	1.84 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.32)	8.43 ($\pm$ 1.43)	5.30 ($\pm$ 0.92)
		0.9	1.84 ($\pm$ 0.06)	1.06 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.10 ($\pm$ 0.34)	8.52 ($\pm$ 1.65)	7.72 ($\pm$ 0.48)
$\mathbf{A}_5(t)$	2000	0.0	6.50 ($\pm$ 0.12)	6.85 ($\pm$ 0.13)	1.83 ($\pm$ 0.73)	1.72 ($\pm$ 0.75)	48.31 ($\pm$ 4.08)	1.99 ($\pm$ 1.43)
		0.3	6.50 ($\pm$ 0.16)	6.81 ($\pm$ 0.17)	2.13 ($\pm$ 0.93)	1.98 ($\pm$ 0.91)	48.25 ($\pm$ 3.58)	17.12 ($\pm$ 2.87)
		0.6	6.50 ($\pm$ 0.16)	6.68 ($\pm$ 0.17)	2.28 ($\pm$ 0.98)	2.12 ($\pm$ 0.96)	48.30 ($\pm$ 3.65)	34.21 ($\pm$ 2.89)
		0.9	6.51 ($\pm$ 0.17)	6.44 ($\pm$ 0.18)	2.41 ($\pm$ 1.01)	2.24 ($\pm$ 1.00)	48.32 ($\pm$ 3.72)	51.32 ($\pm$ 2.89)
	5000	0.0	5.58 ($\pm$ 0.07)	5.93 ($\pm$ 0.08)	1.25 ($\pm$ 0.50)	1.19 ($\pm$ 0.53)	70.38 ($\pm$ 2.85)	1.53 ($\pm$ 0.93)
		0.3	5.55 ($\pm$ 0.08)	5.88 ($\pm$ 0.08)	1.24 ($\pm$ 0.41)	1.19 ($\pm$ 0.38)	70.32 ($\pm$ 2.64)	15.32 ($\pm$ 1.72)
		0.6	5.54 ($\pm$ 0.08)	5.79 ($\pm$ 0.08)	1.32 ($\pm$ 0.43)	1.27 ($\pm$ 0.41)	70.28 ($\pm$ 2.66)	30.68 ($\pm$ 1.72)
		0.9	5.54 ($\pm$ 0.08)	5.64 ($\pm$ 0.08)	1.40 ($\pm$ 0.45)	1.36 ($\pm$ 0.43)	70.25 ($\pm$ 2.73)	46.04 ($\pm$ 1.69)
$\mathbf{A}_6(t)$	2000	0.0	4.75 ($\pm$ 0.09)	4.83 ($\pm$ 0.09)	1.60 ($\pm$ 0.83)	1.62 ($\pm$ 0.83)	11.05 ($\pm$ 1.61)	1.81 ($\pm$ 1.23)
		0.3	4.74 ($\pm$ 0.11)	4.77 ($\pm$ 0.11)	1.79 ($\pm$ 0.71)	1.76 ($\pm$ 0.70)	10.67 ($\pm$ 1.52)	10.17 ($\pm$ 2.79)
		0.6	4.74 ($\pm$ 0.10)	4.58 ($\pm$ 0.11)	1.95 ($\pm$ 0.77)	1.89 ($\pm$ 0.78)	10.77 ($\pm$ 1.69)	20.95 ($\pm$ 2.58)
		0.9	4.74 ($\pm$ 0.10)	4.21 ($\pm$ 0.10)	2.10 ($\pm$ 0.83)	2.02 ($\pm$ 0.83)	10.99 ($\pm$ 1.84)	31.78 ($\pm$ 2.21)
	5000	0.0	3.74 ($\pm$ 0.06)	3.82 ($\pm$ 0.06)	0.96 ($\pm$ 0.42)	0.95 ($\pm$ 0.41)	14.48 ($\pm$ 1.19)	1.11 ($\pm$ 0.83)
		0.3	3.73 ($\pm$ 0.07)	3.78 ($\pm$ 0.07)	0.97 ($\pm$ 0.33)	0.98 ($\pm$ 0.33)	14.50 ($\pm$ 1.39)	8.01 ($\pm$ 1.28)
		0.6	3.73 ($\pm$ 0.07)	3.67 ($\pm$ 0.07)	1.06 ($\pm$ 0.37)	1.06 ($\pm$ 0.35)	14.45 ($\pm$ 1.50)	16.07 ($\pm$ 1.19)
		0.9	3.72 ($\pm$ 0.07)	3.46 ($\pm$ 0.05)	1.15 ($\pm$ 0.40)	1.14 ($\pm$ 0.38)	14.44 ($\pm$ 1.62)	24.11 ($\pm$ 1.09)

Table 6: MAE errors ($\times 10^{-2}$) for Case 2(b) - $\sigma_1^2(t) = 1 + 0.3 \sin(2\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.52 ($\pm$ 0.10)	3.52 ($\pm$ 0.10)	1.54 ($\pm$ 0.66)	1.52 ($\pm$ 0.66)	22.39 ($\pm$ 2.22)	1.96 ($\pm$ 1.43)
		0.3	3.51 ($\pm$ 0.09)	3.43 ($\pm$ 0.09)	1.72 ($\pm$ 0.69)	1.63 ($\pm$ 0.70)	22.33 ($\pm$ 2.07)	6.34 ($\pm$ 2.58)
		0.6	3.51 ($\pm$ 0.10)	3.13 ($\pm$ 0.09)	1.88 ($\pm$ 0.74)	1.76 ($\pm$ 0.77)	22.27 ($\pm$ 2.51)	13.21 ($\pm$ 1.97)
		0.9	3.52 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.02 ($\pm$ 0.80)	1.89 ($\pm$ 0.78)	22.23 ($\pm$ 2.92)	20.14 ($\pm$ 1.26)
	5000	0.0	2.28 ($\pm$ 0.06)	2.28 ($\pm$ 0.06)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.37)	7.11 ($\pm$ 1.15)	1.16 ($\pm$ 0.78)
		0.3	2.24 ($\pm$ 0.06)	2.19 ($\pm$ 0.06)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.29)	7.10 ($\pm$ 1.21)	3.62 ($\pm$ 1.23)
		0.6	2.24 ($\pm$ 0.06)	2.02 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.31)	7.20 ($\pm$ 1.32)	7.03 ($\pm$ 0.99)
		0.9	2.24 ($\pm$ 0.06)	1.64 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.10 ($\pm$ 0.33)	7.36 ($\pm$ 1.50)	10.39 ($\pm$ 0.61)
$\mathbf{A}_2(t)$	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.50 ($\pm$ 0.70)	1.47 ($\pm$ 0.70)	22.57 ($\pm$ 2.21)	1.72 ($\pm$ 1.38)
		0.3	3.46 ($\pm$ 0.10)	3.39 ($\pm$ 0.10)	1.74 ($\pm$ 0.66)	1.64 ($\pm$ 0.69)	22.53 ($\pm$ 2.07)	6.13 ($\pm$ 2.33)
		0.6	3.46 ($\pm$ 0.10)	3.09 ($\pm$ 0.09)	1.91 ($\pm$ 0.71)	1.78 ($\pm$ 0.76)	22.45 ($\pm$ 2.52)	12.87 ($\pm$ 2.01)
		0.9	3.46 ($\pm$ 0.11)	2.39 ($\pm$ 0.08)	2.06 ($\pm$ 0.77)	1.90 ($\pm$ 0.80)	22.39 ($\pm$ 2.98)	19.77 ($\pm$ 1.44)
	5000	0.0	2.22 ($\pm$ 0.05)	2.23 ($\pm$ 0.05)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.38)	6.84 ($\pm$ 1.25)	1.25 ($\pm$ 0.73)
		0.3	2.20 ($\pm$ 0.05)	2.16 ($\pm$ 0.04)	0.91 ($\pm$ 0.31)	0.90 ($\pm$ 0.29)	6.79 ($\pm$ 1.18)	3.56 ($\pm$ 1.18)
		0.6	2.19 ($\pm$ 0.05)	1.97 ($\pm$ 0.04)	1.01 ($\pm$ 0.34)	0.99 ($\pm$ 0.31)	6.90 ($\pm$ 1.29)	6.97 ($\pm$ 0.94)
		0.9	2.19 ($\pm$ 0.05)	1.54 ($\pm$ 0.03)	1.11 ($\pm$ 0.36)	1.07 ($\pm$ 0.32)	7.05 ($\pm$ 1.46)	10.34 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	5.18 ($\pm$ 0.10)	5.30 ($\pm$ 0.10)	1.62 ($\pm$ 0.67)	1.52 ($\pm$ 0.68)	13.67 ($\pm$ 0.75)	1.94 ($\pm$ 1.59)
		0.3	5.18 ($\pm$ 0.10)	5.25 ($\pm$ 0.10)	1.82 ($\pm$ 0.82)	1.77 ($\pm$ 0.88)	13.89 ($\pm$ 1.19)	12.05 ($\pm$ 2.40)
		0.6	5.19 ($\pm$ 0.10)	5.09 ($\pm$ 0.09)	1.96 ($\pm$ 0.90)	1.90 ($\pm$ 0.94)	13.96 ($\pm$ 1.47)	24.07 ($\pm$ 1.92)
		0.9	5.19 ($\pm$ 0.10)	4.76 ($\pm$ 0.09)	2.10 ($\pm$ 0.95)	2.01 ($\pm$ 0.99)	14.05 ($\pm$ 1.80)	36.13 ($\pm$ 1.44)
	5000	0.0	4.24 ($\pm$ 0.06)	4.36 ($\pm$ 0.06)	0.99 ($\pm$ 0.33)	1.01 ($\pm$ 0.36)	22.17 ($\pm$ 1.44)	1.04 ($\pm$ 0.79)
		0.3	4.24 ($\pm$ 0.07)	4.32 ($\pm$ 0.06)	1.00 ($\pm$ 0.41)	1.00 ($\pm$ 0.40)	22.18 ($\pm$ 1.51)	9.80 ($\pm$ 1.27)
		0.6	4.24 ($\pm$ 0.07)	4.22 ($\pm$ 0.06)	1.09 ($\pm$ 0.43)	1.08 ($\pm$ 0.42)	22.13 ($\pm$ 1.65)	19.24 ($\pm$ 1.17)
		0.9	4.24 ($\pm$ 0.06)	4.03 ($\pm$ 0.05)	1.18 ($\pm$ 0.46)	1.16 ($\pm$ 0.45)	22.08 ($\pm$ 1.83)	28.62 ($\pm$ 1.06)
$\mathbf{A}_4(t)$	2000	0.0	3.25 ($\pm$ 0.09)	3.26 ($\pm$ 0.09)	1.49 ($\pm$ 0.68)	1.48 ($\pm$ 0.67)	24.79 ($\pm$ 2.24)	1.84 ($\pm$ 1.49)
		0.3	3.25 ($\pm$ 0.08)	3.15 ($\pm$ 0.08)	1.70 ($\pm$ 0.69)	1.61 ($\pm$ 0.72)	24.76 ($\pm$ 2.11)	5.60 ($\pm$ 2.34)
		0.6	3.25 ($\pm$ 0.09)	2.82 ($\pm$ 0.07)	1.86 ($\pm$ 0.75)	1.75 ($\pm$ 0.78)	24.68 ($\pm$ 2.57)	11.64 ($\pm$ 1.82)
		0.9	3.26 ($\pm$ 0.10)	2.00 ($\pm$ 0.06)	2.01 ($\pm$ 0.80)	1.88 ($\pm$ 0.81)	24.61 ($\pm$ 3.06)	17.83 ($\pm$ 1.03)
	5000	0.0	1.89 ($\pm$ 0.05)	1.89 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.90 ($\pm$ 0.34)	9.48 ($\pm$ 1.31)	1.20 ($\pm$ 0.72)
		0.3	1.86 ($\pm$ 0.05)	1.80 ($\pm$ 0.05)	0.92 ($\pm$ 0.31)	0.91 ($\pm$ 0.29)	9.44 ($\pm$ 1.26)	2.82 ($\pm$ 1.16)
		0.6	1.86 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	9.53 ($\pm$ 1.37)	5.38 ($\pm$ 0.90)
		0.9	1.86 ($\pm$ 0.06)	1.02 ($\pm$ 0.03)	1.11 ($\pm$ 0.37)	1.08 ($\pm$ 0.32)	9.64 ($\pm$ 1.55)	7.89 ($\pm$ 0.50)
$\mathbf{A}_5(t)$	2000	0.0	6.48 ($\pm$ 0.11)	6.82 ($\pm$ 0.13)	1.82 ($\pm$ 0.70)	1.68 ($\pm$ 0.72)	47.92 ($\pm$ 4.14)	1.94 ($\pm$ 1.48)
		0.3	6.48 ($\pm$ 0.15)	6.78 ($\pm$ 0.16)	2.13 ($\pm$ 0.93)	1.92 ($\pm$ 0.86)	47.87 ($\pm$ 3.67)	17.07 ($\pm$ 3.00)
		0.6	6.48 ($\pm$ 0.16)	6.65 ($\pm$ 0.17)	2.28 ($\pm$ 0.96)	2.06 ($\pm$ 0.92)	47.93 ($\pm$ 3.73)	34.17 ($\pm$ 3.00)
		0.9	6.49 ($\pm$ 0.17)	6.42 ($\pm$ 0.18)	2.43 ($\pm$ 1.00)	2.17 ($\pm$ 0.97)	47.99 ($\pm$ 3.79)	51.29 ($\pm$ 2.96)
	5000	0.0	5.56 ($\pm$ 0.07)	5.91 ($\pm$ 0.08)	1.25 ($\pm$ 0.49)	1.17 ($\pm$ 0.53)	71.37 ($\pm$ 2.93)	1.48 ($\pm$ 0.96)
		0.3	5.54 ($\pm$ 0.08)	5.86 ($\pm$ 0.08)	1.22 ($\pm$ 0.41)	1.16 ($\pm$ 0.39)	71.24 ($\pm$ 2.73)	15.33 ($\pm$ 1.71)
		0.6	5.53 ($\pm$ 0.08)	5.78 ($\pm$ 0.08)	1.30 ($\pm$ 0.43)	1.25 ($\pm$ 0.40)	71.21 ($\pm$ 2.74)	30.69 ($\pm$ 1.71)
		0.9	5.53 ($\pm$ 0.08)	5.63 ($\pm$ 0.08)	1.37 ($\pm$ 0.46)	1.33 ($\pm$ 0.42)	71.18 ($\pm$ 2.81)	46.06 ($\pm$ 1.68)
$\mathbf{A}_6(t)$	2000	0.0	4.73 ($\pm$ 0.09)	4.81 ($\pm$ 0.09)	1.61 ($\pm$ 0.80)	1.60 ($\pm$ 0.80)	10.09 ($\pm$ 1.07)	1.84 ($\pm$ 1.17)
		0.3	4.73 ($\pm$ 0.11)	4.76 ($\pm$ 0.11)	1.81 ($\pm$ 0.67)	1.70 ($\pm$ 0.69)	9.95 ($\pm$ 1.14)	10.07 ($\pm$ 2.78)
		0.6	4.73 ($\pm$ 0.10)	4.57 ($\pm$ 0.11)	1.97 ($\pm$ 0.73)	1.82 ($\pm$ 0.77)	10.01 ($\pm$ 1.24)	20.87 ($\pm$ 2.55)
		0.9	4.73 ($\pm$ 0.10)	4.19 ($\pm$ 0.10)	2.11 ($\pm$ 0.79)	1.94 ($\pm$ 0.82)	10.14 ($\pm$ 1.34)	31.74 ($\pm$ 2.20)
	5000	0.0	3.74 ($\pm$ 0.06)	3.81 ($\pm$ 0.06)	0.97 ($\pm$ 0.40)	0.92 ($\pm$ 0.41)	14.68 ($\pm$ 1.16)	1.15 ($\pm$ 0.83)
		0.3	3.72 ($\pm$ 0.07)	3.77 ($\pm$ 0.07)	0.96 ($\pm$ 0.34)	0.97 ($\pm$ 0.33)	14.66 ($\pm$ 1.32)	7.98 ($\pm$ 1.28)
		0.6	3.72 ($\pm$ 0.07)	3.66 ($\pm$ 0.06)	1.05 ($\pm$ 0.38)	1.05 ($\pm$ 0.35)	14.61 ($\pm$ 1.42)	16.05 ($\pm$ 1.20)
		0.9	3.71 ($\pm$ 0.07)	3.46 ($\pm$ 0.05)	1.14 ($\pm$ 0.40)	1.13 ($\pm$ 0.37)	14.59 ($\pm$ 1.54)	24.10 ($\pm$ 1.10)

Table 7: MAE errors ($\times 10^{-2}$) for Case 2(c) - $\sigma_2^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.60 ($\pm$ 0.09)	3.60 ($\pm$ 0.09)	1.51 ($\pm$ 0.68)	1.52 ($\pm$ 0.68)	23.71 ($\pm$ 2.11)	1.91 ($\pm$ 1.38)
		0.3	3.59 ($\pm$ 0.10)	3.50 ($\pm$ 0.10)	1.70 ($\pm$ 0.71)	1.67 ($\pm$ 0.70)	23.57 ($\pm$ 1.96)	6.71 ($\pm$ 2.51)
		0.6	3.59 ($\pm$ 0.10)	3.20 ($\pm$ 0.09)	1.87 ($\pm$ 0.77)	1.81 ($\pm$ 0.76)	23.57 ($\pm$ 2.36)	13.85 ($\pm$ 1.93)
		0.9	3.59 ($\pm$ 0.11)	2.53 ($\pm$ 0.08)	2.02 ($\pm$ 0.82)	1.93 ($\pm$ 0.78)	23.61 ($\pm$ 2.78)	21.00 ($\pm$ 1.23)
	5000	0.0	2.32 ($\pm$ 0.06)	2.31 ($\pm$ 0.06)	0.92 ($\pm$ 0.37)	0.89 ($\pm$ 0.36)	8.74 ($\pm$ 1.05)	1.06 ($\pm$ 0.74)
		0.3	2.29 ($\pm$ 0.06)	2.23 ($\pm$ 0.06)	0.94 ($\pm$ 0.30)	0.93 ($\pm$ 0.29)	8.81 ($\pm$ 1.05)	3.82 ($\pm$ 1.32)
		0.6	2.28 ($\pm$ 0.06)	2.05 ($\pm$ 0.05)	1.04 ($\pm$ 0.33)	1.02 ($\pm$ 0.31)	8.95 ($\pm$ 1.24)	7.32 ($\pm$ 1.06)
		0.9	2.28 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.14 ($\pm$ 0.37)	1.10 ($\pm$ 0.33)	9.11 ($\pm$ 1.49)	10.76 ($\pm$ 0.63)
$\mathbf{A}_2(t)$	2000	0.0	3.56 ($\pm$ 0.10)	3.57 ($\pm$ 0.10)	1.48 ($\pm$ 0.73)	1.49 ($\pm$ 0.72)	23.95 ($\pm$ 2.07)	1.66 ($\pm$ 1.32)
		0.3	3.56 ($\pm$ 0.10)	3.47 ($\pm$ 0.10)	1.73 ($\pm$ 0.69)	1.70 ($\pm$ 0.69)	23.82 ($\pm$ 1.90)	6.48 ($\pm$ 2.41)
		0.6	3.56 ($\pm$ 0.10)	3.17 ($\pm$ 0.09)	1.89 ($\pm$ 0.75)	1.83 ($\pm$ 0.75)	23.85 ($\pm$ 2.25)	13.54 ($\pm$ 1.96)
		0.9	3.56 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.05 ($\pm$ 0.80)	1.95 ($\pm$ 0.80)	23.91 ($\pm$ 2.65)	20.66 ($\pm$ 1.36)
	5000	0.0	2.28 ($\pm$ 0.06)	2.27 ($\pm$ 0.05)	0.92 ($\pm$ 0.38)	0.90 ($\pm$ 0.37)	10.41 ($\pm$ 1.04)	1.15 ($\pm$ 0.71)
		0.3	2.26 ($\pm$ 0.05)	2.20 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.92 ($\pm$ 0.28)	10.45 ($\pm$ 1.07)	3.76 ($\pm$ 1.24)
		0.6	2.26 ($\pm$ 0.05)	2.00 ($\pm$ 0.04)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	10.56 ($\pm$ 1.24)	7.25 ($\pm$ 0.99)
		0.9	2.25 ($\pm$ 0.05)	1.55 ($\pm$ 0.03)	1.12 ($\pm$ 0.36)	1.08 ($\pm$ 0.32)	10.69 ($\pm$ 1.48)	10.69 ($\pm$ 0.62)
$\mathbf{A}_3(t)$	2000	0.0	5.25 ($\pm$ 0.10)	5.38 ($\pm$ 0.10)	1.59 ($\pm$ 0.66)	1.56 ($\pm$ 0.67)	16.13 ($\pm$ 1.18)	1.88 ($\pm$ 1.68)
		0.3	5.25 ($\pm$ 0.10)	5.33 ($\pm$ 0.10)	1.81 ($\pm$ 0.86)	1.79 ($\pm$ 0.86)	16.35 ($\pm$ 1.43)	12.38 ($\pm$ 2.45)
		0.6	5.25 ($\pm$ 0.11)	5.16 ($\pm$ 0.10)	1.95 ($\pm$ 0.93)	1.91 ($\pm$ 0.94)	16.45 ($\pm$ 1.73)	24.68 ($\pm$ 1.97)
		0.9	5.26 ($\pm$ 0.11)	4.83 ($\pm$ 0.10)	2.10 ($\pm$ 0.98)	2.02 ($\pm$ 0.98)	16.55 ($\pm$ 2.03)	36.99 ($\pm$ 1.46)
	5000	0.0	4.28 ($\pm$ 0.06)	4.39 ($\pm$ 0.06)	1.01 ($\pm$ 0.33)	0.99 ($\pm$ 0.35)	22.87 ($\pm$ 1.14)	0.99 ($\pm$ 0.77)
		0.3	4.27 ($\pm$ 0.07)	4.36 ($\pm$ 0.07)	1.00 ($\pm$ 0.41)	1.01 ($\pm$ 0.40)	22.91 ($\pm$ 1.31)	10.02 ($\pm$ 1.30)
		0.6	4.28 ($\pm$ 0.07)	4.25 ($\pm$ 0.06)	1.10 ($\pm$ 0.44)	1.09 ($\pm$ 0.42)	22.88 ($\pm$ 1.45)	19.57 ($\pm$ 1.20)
		0.9	4.28 ($\pm$ 0.06)	4.05 ($\pm$ 0.05)	1.18 ($\pm$ 0.46)	1.17 ($\pm$ 0.44)	22.85 ($\pm$ 1.64)	29.05 ($\pm$ 1.07)
$\mathbf{A}_4(t)$	2000	0.0	3.33 ($\pm$ 0.09)	3.33 ($\pm$ 0.09)	1.48 ($\pm$ 0.70)	1.49 ($\pm$ 0.69)	26.11 ($\pm$ 2.14)	1.81 ($\pm$ 1.44)
		0.3	3.32 ($\pm$ 0.09)	3.23 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.65 ($\pm$ 0.71)	25.99 ($\pm$ 2.02)	5.93 ($\pm$ 2.42)
		0.6	3.32 ($\pm$ 0.09)	2.89 ($\pm$ 0.08)	1.86 ($\pm$ 0.77)	1.79 ($\pm$ 0.78)	25.97 ($\pm$ 2.46)	12.28 ($\pm$ 1.82)
		0.9	3.33 ($\pm$ 0.11)	2.06 ($\pm$ 0.07)	2.01 ($\pm$ 0.82)	1.91 ($\pm$ 0.80)	25.98 ($\pm$ 2.93)	18.68 ($\pm$ 1.01)
	5000	0.0	1.93 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.92 ($\pm$ 0.34)	0.89 ($\pm$ 0.33)	10.46 ($\pm$ 1.17)	1.09 ($\pm$ 0.70)
		0.3	1.90 ($\pm$ 0.05)	1.83 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	10.47 ($\pm$ 1.14)	3.03 ($\pm$ 1.25)
		0.6	1.90 ($\pm$ 0.06)	1.60 ($\pm$ 0.05)	1.03 ($\pm$ 0.34)	1.01 ($\pm$ 0.31)	10.59 ($\pm$ 1.30)	5.68 ($\pm$ 0.96)
		0.9	1.90 ($\pm$ 0.06)	1.04 ($\pm$ 0.03)	1.13 ($\pm$ 0.37)	1.09 ($\pm$ 0.33)	10.72 ($\pm$ 1.54)	8.25 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	6.57 ($\pm$ 0.11)	6.93 ($\pm$ 0.13)	1.81 ($\pm$ 0.73)	1.71 ($\pm$ 0.76)	51.66 ($\pm$ 3.96)	1.95 ($\pm$ 1.39)
		0.3	6.57 ($\pm$ 0.16)	6.88 ($\pm$ 0.17)	2.14 ($\pm$ 0.94)	1.98 ($\pm$ 0.91)	51.69 ($\pm$ 3.46)	17.42 ($\pm$ 2.83)
		0.6	6.57 ($\pm$ 0.16)	6.76 ($\pm$ 0.17)	2.29 ($\pm$ 0.98)	2.12 ($\pm$ 0.96)	51.70 ($\pm$ 3.49)	34.76 ($\pm$ 2.85)
		0.9	6.58 ($\pm$ 0.17)	6.52 ($\pm$ 0.18)	2.43 ($\pm$ 1.02)	2.24 ($\pm$ 1.01)	51.70 ($\pm$ 3.52)	52.10 ($\pm$ 2.85)
	5000	0.0	5.61 ($\pm$ 0.07)	5.96 ($\pm$ 0.08)	1.27 ($\pm$ 0.51)	1.19 ($\pm$ 0.53)	71.18 ($\pm$ 2.79)	1.50 ($\pm$ 0.98)
		0.3	5.58 ($\pm$ 0.08)	5.91 ($\pm$ 0.09)	1.23 ($\pm$ 0.40)	1.20 ($\pm$ 0.39)	71.10 ($\pm$ 2.66)	15.45 ($\pm$ 1.68)
		0.6	5.58 ($\pm$ 0.08)	5.82 ($\pm$ 0.08)	1.31 ($\pm$ 0.41)	1.29 ($\pm$ 0.41)	71.06 ($\pm$ 2.68)	30.91 ($\pm$ 1.71)
		0.9	5.58 ($\pm$ 0.08)	5.66 ($\pm$ 0.08)	1.39 ($\pm$ 0.44)	1.37 ($\pm$ 0.43)	71.03 ($\pm$ 2.75)	46.37 ($\pm$ 1.70)
$\mathbf{A}_6(t)$	2000	0.0	4.79 ($\pm$ 0.09)	4.87 ($\pm$ 0.09)	1.58 ($\pm$ 0.83)	1.59 ($\pm$ 0.81)	12.19 ($\pm$ 1.30)	1.85 ($\pm$ 1.08)
		0.3	4.78 ($\pm$ 0.11)	4.81 ($\pm$ 0.11)	1.79 ($\pm$ 0.70)	1.73 ($\pm$ 0.70)	11.89 ($\pm$ 1.27)	10.47 ($\pm$ 2.74)
		0.6	4.78 ($\pm$ 0.10)	4.62 ($\pm$ 0.11)	1.95 ($\pm$ 0.77)	1.86 ($\pm$ 0.78)	12.03 ($\pm$ 1.44)	21.54 ($\pm$ 2.52)
		0.9	4.78 ($\pm$ 0.10)	4.24 ($\pm$ 0.10)	2.10 ($\pm$ 0.82)	1.98 ($\pm$ 0.83)	12.27 ($\pm$ 1.63)	32.64 ($\pm$ 2.14)
	5000	0.0	3.76 ($\pm$ 0.06)	3.83 ($\pm$ 0.06)	0.97 ($\pm$ 0.42)	0.92 ($\pm$ 0.40)	13.85 ($\pm$ 1.16)	1.11 ($\pm$ 0.81)
		0.3	3.74 ($\pm$ 0.07)	3.78 ($\pm$ 0.07)	0.96 ($\pm$ 0.34)	0.97 ($\pm$ 0.32)	13.90 ($\pm$ 1.41)	8.16 ($\pm$ 1.31)
		0.6	3.74 ($\pm$ 0.07)	3.67 ($\pm$ 0.07)	1.05 ($\pm$ 0.37)	1.05 ($\pm$ 0.34)	13.89 ($\pm$ 1.50)	16.33 ($\pm$ 1.21)
		0.9	3.73 ($\pm$ 0.07)	3.46 ($\pm$ 0.05)	1.14 ($\pm$ 0.41)	1.13 ($\pm$ 0.36)	13.92 ($\pm$ 1.60)	24.47 ($\pm$ 1.08)

Table 8: MAE errors ($\times 10^{-2}$) for Case 2(d) - $\sigma_3^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

function	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
$\mathbf{A}_1(t)$	2000	0.0	3.03 ($\pm$ 0.09)	3.04 ($\pm$ 0.09)	1.38 ($\pm$ 0.59)	1.29 ($\pm$ 0.55)	15.01 ($\pm$ 1.82)	2.13 ($\pm$ 1.48)
		0.3	3.02 ($\pm$ 0.08)	2.96 ($\pm$ 0.08)	1.48 ($\pm$ 0.61)	1.41 ($\pm$ 0.60)	14.82 ($\pm$ 1.76)	6.41 ($\pm$ 2.66)
		0.6	3.03 ($\pm$ 0.09)	2.72 ($\pm$ 0.08)	1.61 ($\pm$ 0.68)	1.53 ($\pm$ 0.64)	14.76 ($\pm$ 2.15)	13.17 ($\pm$ 2.18)
		0.9	3.03 ($\pm$ 0.09)	2.19 ($\pm$ 0.07)	1.74 ($\pm$ 0.74)	1.64 ($\pm$ 0.65)	14.73 ($\pm$ 2.57)	20.00 ($\pm$ 1.39)
	5000	0.0	2.03 ($\pm$ 0.05)	2.04 ($\pm$ 0.05)	0.78 ($\pm$ 0.30)	0.77 ($\pm$ 0.32)	3.64 ($\pm$ 0.97)	1.21 ($\pm$ 0.88)
		0.3	2.00 ($\pm$ 0.05)	1.98 ($\pm$ 0.05)	0.82 ($\pm$ 0.26)	0.79 ($\pm$ 0.25)	3.57 ($\pm$ 0.92)	3.76 ($\pm$ 1.33)
		0.6	2.00 ($\pm$ 0.05)	1.84 ($\pm$ 0.05)	0.91 ($\pm$ 0.29)	0.86 ($\pm$ 0.27)	3.67 ($\pm$ 0.98)	7.37 ($\pm$ 1.07)
		0.9	2.00 ($\pm$ 0.05)	1.57 ($\pm$ 0.03)	0.99 ($\pm$ 0.31)	0.93 ($\pm$ 0.29)	3.81 ($\pm$ 1.07)	10.93 ($\pm$ 0.66)
$\mathbf{A}_2(t)$	2000	0.0	2.98 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.33 ($\pm$ 0.62)	1.24 ($\pm$ 0.61)	15.17 ($\pm$ 1.84)	1.86 ($\pm$ 1.54)
		0.3	2.98 ($\pm$ 0.08)	2.92 ($\pm$ 0.09)	1.50 ($\pm$ 0.59)	1.43 ($\pm$ 0.59)	15.01 ($\pm$ 1.79)	6.22 ($\pm$ 2.33)
		0.6	2.98 ($\pm$ 0.09)	2.68 ($\pm$ 0.08)	1.64 ($\pm$ 0.64)	1.54 ($\pm$ 0.65)	14.96 ($\pm$ 2.16)	12.81 ($\pm$ 2.16)
		0.9	2.98 ($\pm$ 0.09)	2.11 ($\pm$ 0.07)	1.78 ($\pm$ 0.70)	1.65 ($\pm$ 0.66)	14.92 ($\pm$ 2.56)	19.60 ($\pm$ 1.53)
	5000	0.0	1.98 ($\pm$ 0.04)	1.99 ($\pm$ 0.04)	0.78 ($\pm$ 0.31)	0.77 ($\pm$ 0.33)	4.30 ($\pm$ 0.79)	1.32 ($\pm$ 0.69)
		0.3	1.96 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.76 ($\pm$ 0.26)	4.23 ($\pm$ 0.81)	3.70 ($\pm$ 1.30)
		0.6	1.95 ($\pm$ 0.04)	1.78 ($\pm$ 0.04)	0.90 ($\pm$ 0.27)	0.83 ($\pm$ 0.28)	4.30 ($\pm$ 0.88)	7.30 ($\pm$ 1.05)
		0.9	1.95 ($\pm$ 0.05)	1.44 ($\pm$ 0.03)	0.98 ($\pm$ 0.30)	0.89 ($\pm$ 0.29)	4.41 ($\pm$ 0.98)	10.86 ($\pm$ 0.69)
$\mathbf{A}_3(t)$	2000	0.0	4.80 ($\pm$ 0.09)	4.95 ($\pm$ 0.09)	1.45 ($\pm$ 0.60)	1.32 ($\pm$ 0.56)	12.45 ($\pm$ 1.05)	1.93 ($\pm$ 1.68)
		0.3	4.80 ($\pm$ 0.09)	4.91 ($\pm$ 0.09)	1.62 ($\pm$ 0.72)	1.53 ($\pm$ 0.76)	12.41 ($\pm$ 1.21)	13.19 ($\pm$ 2.56)
		0.6	4.80 ($\pm$ 0.09)	4.78 ($\pm$ 0.09)	1.75 ($\pm$ 0.78)	1.64 ($\pm$ 0.82)	12.43 ($\pm$ 1.45)	26.26 ($\pm$ 2.11)
		0.9	4.81 ($\pm$ 0.09)	4.55 ($\pm$ 0.08)	1.86 ($\pm$ 0.83)	1.73 ($\pm$ 0.87)	12.49 ($\pm$ 1.76)	39.33 ($\pm$ 1.62)
	5000	0.0	4.07 ($\pm$ 0.06)	4.23 ($\pm$ 0.06)	0.88 ($\pm$ 0.31)	0.87 ($\pm$ 0.32)	24.04 ($\pm$ 1.18)	1.01 ($\pm$ 0.73)
		0.3	4.07 ($\pm$ 0.06)	4.20 ($\pm$ 0.06)	0.89 ($\pm$ 0.35)	0.87 ($\pm$ 0.36)	24.10 ($\pm$ 1.26)	11.27 ($\pm$ 1.34)
		0.6	4.07 ($\pm$ 0.06)	4.13 ($\pm$ 0.05)	0.97 ($\pm$ 0.37)	0.95 ($\pm$ 0.38)	24.04 ($\pm$ 1.34)	22.19 ($\pm$ 1.27)
		0.9	4.08 ($\pm$ 0.06)	4.00 ($\pm$ 0.05)	1.04 ($\pm$ 0.40)	1.01 ($\pm$ 0.40)	23.99 ($\pm$ 1.45)	33.06 ($\pm$ 1.17)
$\mathbf{A}_4(t)$	2000	0.0	2.75 ($\pm$ 0.08)	2.75 ($\pm$ 0.08)	1.34 ($\pm$ 0.60)	1.24 ($\pm$ 0.56)	17.41 ($\pm$ 1.85)	1.98 ($\pm$ 1.59)
		0.3	2.74 ($\pm$ 0.07)	2.66 ($\pm$ 0.07)	1.46 ($\pm$ 0.61)	1.40 ($\pm$ 0.62)	17.24 ($\pm$ 1.81)	5.47 ($\pm$ 2.38)
		0.6	2.75 ($\pm$ 0.08)	2.38 ($\pm$ 0.06)	1.60 ($\pm$ 0.68)	1.52 ($\pm$ 0.66)	17.19 ($\pm$ 2.18)	11.14 ($\pm$ 2.00)
		0.9	2.75 ($\pm$ 0.09)	1.69 ($\pm$ 0.05)	1.74 ($\pm$ 0.73)	1.62 ($\pm$ 0.67)	17.15 ($\pm$ 2.60)	16.99 ($\pm$ 1.07)
	5000	0.0	1.62 ($\pm$ 0.04)	1.62 ($\pm$ 0.04)	0.78 ($\pm$ 0.28)	0.77 ($\pm$ 0.30)	6.40 ($\pm$ 1.03)	1.20 ($\pm$ 0.78)
		0.3	1.60 ($\pm$ 0.05)	1.55 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.77 ($\pm$ 0.26)	6.33 ($\pm$ 0.98)	2.74 ($\pm$ 1.19)
		0.6	1.60 ($\pm$ 0.05)	1.36 ($\pm$ 0.04)	0.90 ($\pm$ 0.28)	0.85 ($\pm$ 0.28)	6.39 ($\pm$ 1.06)	5.22 ($\pm$ 0.96)
		0.9	1.60 ($\pm$ 0.05)	0.91 ($\pm$ 0.03)	0.98 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	6.46 ($\pm$ 1.21)	7.66 ($\pm$ 0.51)
$\mathbf{A}_5(t)$	2000	0.0	6.14 ($\pm$ 0.11)	6.50 ($\pm$ 0.13)	1.69 ($\pm$ 0.67)	1.49 ($\pm$ 0.64)	49.00 ($\pm$ 3.96)	2.13 ($\pm$ 1.46)
		0.3	6.14 ($\pm$ 0.14)	6.47 ($\pm$ 0.16)	1.90 ($\pm$ 0.84)	1.69 ($\pm$ 0.79)	48.91 ($\pm$ 3.46)	18.63 ($\pm$ 2.96)
		0.6	6.14 ($\pm$ 0.15)	6.38 ($\pm$ 0.16)	2.02 ($\pm$ 0.88)	1.81 ($\pm$ 0.83)	48.94 ($\pm$ 3.56)	37.15 ($\pm$ 3.03)
		0.9	6.14 ($\pm$ 0.16)	6.22 ($\pm$ 0.17)	2.14 ($\pm$ 0.91)	1.91 ($\pm$ 0.87)	48.95 ($\pm$ 3.67)	55.70 ($\pm$ 3.07)
	5000	0.0	5.41 ($\pm$ 0.07)	5.80 ($\pm$ 0.08)	1.14 ($\pm$ 0.44)	1.04 ($\pm$ 0.47)	72.32 ($\pm$ 2.64)	1.61 ($\pm$ 1.02)
		0.3	5.39 ($\pm$ 0.08)	5.76 ($\pm$ 0.08)	1.16 ($\pm$ 0.42)	1.03 ($\pm$ 0.35)	72.34 ($\pm$ 2.42)	17.13 ($\pm$ 1.88)
		0.6	5.38 ($\pm$ 0.07)	5.70 ($\pm$ 0.08)	1.23 ($\pm$ 0.43)	1.10 ($\pm$ 0.36)	72.30 ($\pm$ 2.40)	34.35 ($\pm$ 1.86)
		0.9	5.38 ($\pm$ 0.07)	5.60 ($\pm$ 0.08)	1.30 ($\pm$ 0.45)	1.17 ($\pm$ 0.37)	72.26 ($\pm$ 2.44)	51.58 ($\pm$ 1.82)
$\mathbf{A}_6(t)$	2000	0.0	4.41 ($\pm$ 0.08)	4.49 ($\pm$ 0.08)	1.44 ($\pm$ 0.73)	1.43 ($\pm$ 0.73)	12.00 ($\pm$ 1.12)	1.88 ($\pm$ 1.35)
		0.3	4.40 ($\pm$ 0.09)	4.44 ($\pm$ 0.10)	1.57 ($\pm$ 0.62)	1.53 ($\pm$ 0.59)	11.97 ($\pm$ 1.37)	11.07 ($\pm$ 2.88)
		0.6	4.40 ($\pm$ 0.09)	4.30 ($\pm$ 0.10)	1.71 ($\pm$ 0.68)	1.64 ($\pm$ 0.66)	11.98 ($\pm$ 1.47)	22.70 ($\pm$ 2.74)
		0.9	4.40 ($\pm$ 0.09)	4.02 ($\pm$ 0.09)	1.84 ($\pm$ 0.74)	1.73 ($\pm$ 0.71)	12.05 ($\pm$ 1.59)	34.38 ($\pm$ 2.46)
	5000	0.0	3.60 ($\pm$ 0.05)	3.69 ($\pm$ 0.05)	0.84 ($\pm$ 0.36)	0.83 ($\pm$ 0.37)	18.62 ($\pm$ 0.93)	1.15 ($\pm$ 0.87)
		0.3	3.59 ($\pm$ 0.06)	3.66 ($\pm$ 0.06)	0.87 ($\pm$ 0.30)	0.86 ($\pm$ 0.30)	18.61 ($\pm$ 1.07)	9.18 ($\pm$ 1.34)
		0.6	3.59 ($\pm$ 0.06)	3.58 ($\pm$ 0.06)	0.94 ($\pm$ 0.33)	0.92 ($\pm$ 0.32)	18.56 ($\pm$ 1.16)	18.51 ($\pm$ 1.29)
		0.9	3.58 ($\pm$ 0.06)	3.43 ($\pm$ 0.05)	1.02 ($\pm$ 0.35)	0.99 ($\pm$ 0.35)	18.53 ($\pm$ 1.26)	27.82 ($\pm$ 1.22)

两个发现：

- 在配对算法 $\mathbf{A}^*$ 与 iid 算法 $\mathbf{A}^\dagger$ 的比较上： $\mathbf{A}_1(t)$ 、 $\mathbf{A}_2(t)$ 、 $\mathbf{A}_4(t)$ 在 $\rho \geq 0.3$ 时，配对算法就已经超越 iid 算法；而 $\mathbf{A}_3(t)$ 、 $\mathbf{A}_5(t)$ 、 $\mathbf{A}_6(t)$ 通常需要更大的 ρ ，其中 $\mathbf{A}_5(t)$ 的改善最弱。
- 在 $\sigma^2(t)$ 的估计上： $\mathbf{A}_3(t)$ 、 $\mathbf{A}_5(t)$ 、 $\mathbf{A}_6(t)$ 在 $n = 2000 \rightarrow 5000$ 的过程中，方差的估计变得更差。

根本原因（猜测）是固定窗宽 h 的设置下，复杂函数 $\mathbf{A}_3(t)$ 、 $\mathbf{A}_5(t)$ 、 $\mathbf{A}_6(t)$ 的 iid 估计本身就产生了较大误差，而这些误差被带进了 $\sigma^2(t)$ 的估计，从而最终影响配对算法的估计。所以才导致 $\sigma^2(t)$ 差，且增大样本量并没有任何改善。